

NND Schemes and Their Applications to Numerical Simulation of Two- and Three-Dimensional Flows*

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I. Introduction

In the calculation of complex flow fields containing shock waves, the most attention has been paid to the shock-capturing method. In order to capture shock waves smoothly without spurious oscillations near or in the shock regions, mixed dissipative schemes containing free parameters with first-order accuracy near shocks and second-order schemes elsewhere have

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been widely used (see Pulliam, 1985; Jameson and Yoon, 1985; and Zhang *et al.*, 1983, and Zheng, 1986). There are inherent disadvantages in employing these schemes. First, the free parameters are basically determined through numerical experiments; second, the resolution of shock is not very satisfactory. Naturally, the development of nonoscillatory dissipative schemes containing no free parameters with high resolution has been much emphasized recently, such as TVD schemes (see Harten, 1984; van Leer, 1979; Chakravarthy and Osher, 1985; Davis, 1984; and Yee, 1987) and ENO schemes (see Harten, 1986).

In Zhang (1984), Zhang found with a linearized analysis of one-dimensional Navier–Stokes equations that when a proper manipulation of the coefficients of third-order derivatives ν_3 in the corresponding modified differential equations is used, the spurious oscillation at both upstream and downstream shocks can be totally suppressed. The choice of ν_3 must be in accordance with the second law of thermodynamics. The numerical experiments with one-dimensional nonlinear Navier–Stokes equations confirm that the conclusion reached by linearized analysis is valid in general. The importance of the role of ν_3 was not realized before. While it was usually thought that the third-order dispersion term in the modified differential equations would not play an essential role in a linear problem, e.g., in the stability analysis, etc., they do contribute through nonlinear interactions to the unwanted wave structure, and even lead to chaos if they are not properly treated.

Based on these physical concepts, we suggest a new NND scheme (see Zhang, 1988) that is nonoscillatory, contains no free parameters, and is dissipative. The details will be described in this paper. In order to test the effectiveness of the scheme, we have carried out numerical simulations for two- and three-dimensional flows with the Euler equations and Navier–Stokes equations. The results of calculations are given. Finally, some concluding remarks are given.

II. The Importance of the Role of the Third-Order Dispersion Term

A. ANALYTIC ANALYSIS (see Zhang, 1984)

Now we study the problem of one-dimensional shock wave using a time-dependent method. The model equation and boundary conditions are

$$\begin{cases} \frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} - \nu \frac{\partial^2 u}{\partial x^2} = 0, \\ x \rightarrow -\infty, \quad u \rightarrow u_\infty, \quad \frac{\partial u}{\partial x} \rightarrow 0, \end{cases} \quad (2.1)$$

where

$$\begin{aligned} a &= \frac{\gamma + 1}{2\gamma} u_\infty \left(1 - \frac{u_2 u_\infty}{u^2} \right), \\ \nu &= \frac{4}{3} \frac{\mu}{\rho_\infty}, \\ u_2 &= \frac{\gamma - 1}{\gamma + 1} \left(1 + \frac{2}{\gamma - 1} \frac{1}{M_\infty^2} \right) u_\infty. \end{aligned} \quad (2.2)$$

ρ , u , and μ are the fluid density, velocity, and the coefficient of viscosity, respectively. The variable t represents the time and x represents a coordinate (Fig. 1). Free-stream conditions are denoted with the sign “ ∞ .” M_∞ is the free-stream Mach number and γ is the ratio of the specific heat. If the flow is steady and the coefficient of viscosity is constant, Eq. (2.1) represents the exact Navier–Stokes equations governing a normal shock wave.

If a finite-difference method is applied to solving Eq. (2.1), the following modified differential equation, which is actually a finite-difference equation, can be written as

$$\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} - \nu \frac{\partial^2 u}{\partial x^2} = \nu_2 \frac{\partial^2 u}{\partial x^2} + \nu_3 \frac{\partial^3 u}{\partial x^3} - \nu_4 \frac{\partial^4 u}{\partial x^4} + \dots, \quad (2.3)$$

where the right-hand side represents the truncation error for the difference method. For a second-order accurate difference method, $\nu_2 = 0$.

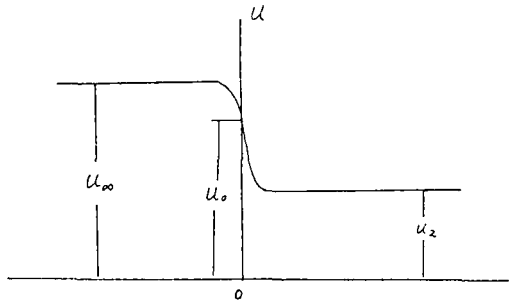


FIG. 1. One dimensional shock wave.

Now, we study the effects of coefficients $\nu_2, \nu_3, \dots$ of the preceding steady solution. In fact, if the flow is steady, Eq. (2.3) can be integrated once to obtain

$$(\nu + \nu_2) \frac{\partial \bar{u}}{\partial x} + \nu_3 \frac{\partial^2 \bar{u}}{\partial x^2} - \nu_4 \frac{\partial^3 \bar{u}}{\partial x^3} = \frac{\gamma + 1}{2\gamma} u_\infty \frac{(\bar{u} - 1)(\bar{u} - \bar{u}_2)}{\bar{u}}, \quad (2.4)$$

where

$$\bar{u} = \frac{u}{u_\infty},$$

$$\bar{u}_2 = \frac{\gamma - 1}{\gamma + 1} \left(1 + \frac{2}{\gamma - 1} \frac{1}{M_\infty^2} \right).$$

Obviously, $\bar{u} = 1$ is the nondimensional velocity in the upstream region of the inviscid shock, and $\bar{u} = \bar{u}_2$ is the nondimensional velocity in the downstream region.

We assume that the numerical oscillations induced by the truncation error of the difference method are small; then, in the upstream region of the shock,

$$\bar{u} = 1 + u', \quad (2.5a)$$

and in the downstream region of the shock,

$$\bar{u} = \bar{u}_2 + u', \quad (2.5b)$$

where $u' \ll 1$ upstream and $u' \ll \bar{u}_2$ downstream. Substituting (2.5a) and (2.5b) into Eq. (2.4) and neglecting the high-order small quantity, we obtain

$$\begin{aligned} \nu_1 \frac{\partial u'}{\partial x} + \nu_3 \frac{\partial^2 u'}{\partial x^2} - \nu_4 \frac{\partial^3 u'}{\partial x^3} &= k_1 u' & (\text{upstream}), \\ \nu_1 \frac{\partial u'}{\partial x} + \nu_3 \frac{\partial^2 u'}{\partial x^2} - \nu_4 \frac{\partial^3 u'}{\partial x^3} &= -k_2 u' & (\text{downstream}), \end{aligned} \quad (2.6)$$

where

$$\begin{aligned} \nu_1 &= \nu + \nu_2, \\ k_1 &= \frac{\gamma + 1}{2\gamma} u_\infty (1 - \bar{u}_2) > 0, \\ k_2 &= \frac{\gamma + 1}{2\gamma} u_\infty \frac{1 - \bar{u}_2}{\bar{u}_2} > 0. \end{aligned}$$

Equation (2.6) is linear and its solution can be determined by following characteristic equations:

$$v_4 \lambda^3 - v_3 \lambda^2 - v_1 \lambda + k_1 = 0 \quad (\text{upstream}), \quad (2.7)$$

$$v_4 \lambda^3 - v_3 \lambda^2 - v_1 \lambda - k_2 = 0 \quad (\text{downstream}). \quad (2.8)$$

To discuss following cases is of some help:

Case 1. $v_1 > 0$, v_3 and v_4 are very small. This case corresponds to using the first-order difference scheme. The solution of Eq. (2.6) is

$$u = 1 + u' = \begin{cases} 1 + Ae^{(k_1/v_1)x} & (\text{upstream } x > 0), \\ \bar{u}_2 + Be^{-(k_2/v_1)x} & (\text{downstream } x > 0); \end{cases} \quad (2.9)$$

A, B are constants. This result shows that there is no numerical oscillation in both sides of the shock.

Case 2. $v_2 = 0$, $v_3 < 0$, v_4 is very small. For inviscid flow ($\nu = 0$) or the flow at high Reynolds number (ν is very small), the solution of Eq. (2.6) is

$$\bar{u} = \begin{cases} 1 + A_1 \exp\left(\frac{v_1}{2|v_3|}x\right) \cos\left(\frac{1}{2|v_3|} \sqrt{4|v_3|k_1 - v_1^2}x\right) \\ \quad + A_2 \exp\left(\frac{v_1}{2|v_3|}x\right) \sin\left(\frac{1}{2|v_3|} \sqrt{4|v_3|k_1 - v_1^2}x\right) & (\text{upstream}), \\ \bar{u}_2 + A \exp\left\{\left[\frac{v_1}{2|v_3|} - \frac{1}{2|v_3|} (v_1^2 + 4k_2|v_3|)^{1/2}\right]x\right\} & (\text{downstream}). \end{cases} \quad (2.10)$$

It is very clear that the spurious oscillations occur in the upstream region of the shock, but not in the downstream region.

Case 3. $v_2 = 0$, $v_3 > 0$, v_4 is very small. In a similar way, we can prove that the spurious oscillations occur in the downstream region of the shock, but not in the upstream region.

Cases 2 and 3 correspond to using the second-order difference scheme. Through the preceding study for one-dimensional Navier–Stokes equations, it is found that the spurious oscillations occurring near the shock with the second-order finite difference equations are related to the dispersion term in the corresponding modified differential equations. If we can keep $v_3 > 0$ in

the upstream region of the shock and $v_3 < 0$ downstream, we may have a smooth shock transition, i.e., the undesirable oscillations can be totally suppressed.

B. PHYSICAL DISCUSSION AND NUMERICAL SIMULATION

(see Zhang and Mao, 1987, and Zhang, 1988)

The preceding conclusion can be also verified by following physical discussion from the second law of thermodynamics. In fact, the one-dimensional Navier-Stokes equations modified by the addition of dispersion terms with coefficient v_3 are as follows:

$$\begin{aligned}\frac{\partial \rho}{\partial t} + \frac{\partial \rho u}{\partial x} &= 0, \\ \rho \frac{\partial u}{\partial t} + \rho u \frac{\partial u}{\partial x} + \frac{\partial p}{\partial x} &= \frac{\partial}{\partial x} \left(\frac{4}{3} \mu \frac{\partial u}{\partial x} \right) + \frac{\partial}{\partial x} \left(v_3 \mu \frac{\partial^2 u}{\partial x^2} \right), \\ \rho \frac{\partial H}{\partial t} - \frac{\partial p}{\partial t} + \rho u \frac{\partial H}{\partial x} &= \frac{\partial}{\partial x} \left(\frac{4}{3} \mu \frac{\partial H}{\partial x} \right) + \frac{\partial}{\partial x} \left(v_3 \mu \frac{\partial^2 H}{\partial x^2} \right),\end{aligned}\quad (2.11)$$

where p represents the pressure, H represents the total enthalpy, and the Prandtl number is assumed to be $3/4$. From Eq. (2.11), we may obtain an equation of entropy s for the heat-isolated system

$$\rho T \frac{Ds}{Dt} = \frac{4}{3} \mu \left(\frac{\partial u}{\partial x} \right)^2 + 3v_3 \left(\frac{\partial u}{\partial x} \right) \left(\frac{\partial^2 u}{\partial x^2} \right). \quad (2.12)$$

Here, Ds/Dt is the substantial derivative of entropy. For a true physical shock, we have, in the upstream of the shock,

$$\frac{\partial u}{\partial x} < 0, \quad \frac{\partial^2 u}{\partial x^2} < 0; \quad (2.13)$$

hence,

$$\left(\frac{\partial u}{\partial x} \right) \left(\frac{\partial^2 u}{\partial x^2} \right) > 0. \quad (2.14)$$

And in the downstream of the shock,

$$\frac{\partial u}{\partial x} < 0, \quad \frac{\partial^2 u}{\partial x^2} > 0; \quad (2.15)$$

hence,

$$\left(\frac{\partial u}{\partial x}\right)\left(\frac{\partial^2 u}{\partial x^2}\right) < 0. \quad (2.16)$$

For inviscid flow or flow at high Reynolds numbers, we may disregard the viscous contribution to the entropy; therefore, we may observe that if $v_3 > 0$ in the whole shock region, then the increasing-entropy condition is met in upstream region but not in the downstream region, and this is associated with the appearance downstream of spurious oscillations; on the other hand, if $v_3 < 0$ in the whole region, then the increasing-entropy condition is not met in the upstream, and there occur spurious oscillations upstream. Now, it is obvious that if we can keep $v_3 > 0$ upstream and $v_3 < 0$ downstream, we may have a smooth shock transition.

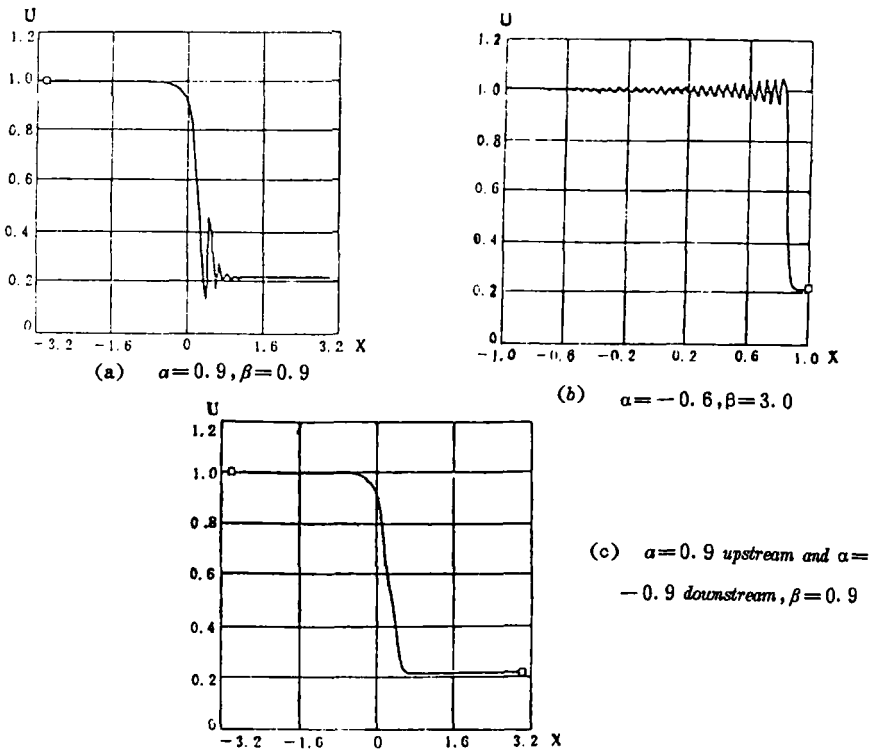


FIG. 2 Numerical simulation for normal shock wave with (2.11).

$$a = 2\gamma v_3 / [(\gamma + 1)\rho_\infty u_\infty \Delta x^2]; \quad \beta = 3\gamma\mu / [8(\gamma + 1)\rho_\infty u_\infty \Delta x]; \quad M_\infty = 4.$$

To test the effectiveness of previous conclusion, we calculated one-dimensional flow with Eqs. (2.11). Numerical results (see Zhang and Mao, 1987) verify the preceding conclusion. Figure 2a shows the case for $v_3 > 0$ in the whole shock region. Figure 2b is for $v_3 < 0$ in the whole shock region. Figure 2c corresponds with $v_3 > 0$ upstream and $v_3 < 0$ downstream. In passing, we may note that there might be individual points in the shock region where $(\partial u / \partial x)(\partial^2 u / \partial x^2) = 0$, and hence when $\mu = 0$, the increasing-entropy condition is not satisfied at these points.

It might be argued that in the limit of mesh size approaching zero, the coefficients v_2 , v_3 , and v_4 , etc., will become zero anyway, and the solution obtained will seem to converge to the required solution of Navier-Stokes equations. But care must be taken that the correct physical solution can only be arrived from a sequence of solutions that satisfies the second law of thermodynamics. Of course, entropy at certain places may decrease if heat conduction is included; nevertheless, the condition imposed here is necessary.

III. A Formulation of the Semi-discretized NND Scheme

(see Zhang, 1988)

A. THE CASE FOR A SCALAR EQUATION

To construct a semi-discretized NND scheme, we start with a one-dimensional scalar equation

$$\frac{\partial u}{\partial t} + \frac{\partial f(u)}{\partial x} = 0. \quad (3.1)$$

Here, $f = au$ and $a = \partial f / \partial u$, a is the characteristic speed, and we may write

$$a = a^+ + a^-, \quad (3.2)$$

where

$$a^+ = (a + |a|)/2, \quad a^- = (a - |a|)/2. \quad (3.3)$$

Define

$$f^+ = a^+ u \quad \text{and} \quad f^- = a^- u; \quad (3.4)$$

we have

$$f = f^+ + f^-. \quad (3.5)$$

Equation (3.1) may be written as

$$\frac{\partial u}{\partial t} + \frac{\partial f^+}{\partial x} + \frac{\partial f^-}{\partial x} = 0. \quad (3.6)$$

It is easy to verify that if the second-order upwind difference scheme is used to evaluate the term $\partial f^+/\partial x$, the coefficient of the third-order derivative in the right-hand side of the modified equation is positive and the coefficient of the fourth-order dissipative term is negative; if the second-order central difference is used, the coefficient of the third-order derivative is negative and the coefficient of the fourth-order dissipative term is zero. Similarly, if the second-order upwind scheme is used to evaluate $\partial f^-/\partial x$, the corresponding coefficient of third-order derivative is negative, while we still have the coefficient of fourth-order dissipative term negative; if the second-order central difference is used, the coefficient of third-order derivative is positive and the coefficient of fourth-order dissipative term remains zero. Based on those findings, we may employ the following difference schemes to evaluate space derivatives.

In the upstream region of a shock:

second-order upwind difference can be used to replace $\partial f^+/\partial x$;
second-order central difference can be used to replace $\partial f^-/\partial x$.

In the downstream region of a shock:

second-order central difference can be used to replace $\partial f^+/\partial x$;
second-order upwind difference can be used to replace $\partial f^-/\partial x$.

In so doing, we have made the proper choice of the sign of the coefficient of the third derivative in the modified differential equation, i.e., v_3 is positive upstream of a shock and v_3 is negative downstream of a shock, and this will provide us with a sharp transition without spurious oscillations, both upstream and downstream of shocks. At the same time, we notice that the coefficient of fourth-order dissipative derivative is negative in the entire region, and that this helps to suppress odd-even decoupling in smooth regions of physical flows.

Now we write down the semi-discretized difference form of Eq. (3.6) as follows: in the upstream region of a shock,

$$\left(\frac{\partial u}{\partial t}\right)_j^n = -\frac{(3f_j^{+n} - 4f_{j-1}^{+n} + f_{j-2}^{+n})}{(2\Delta x)} - \frac{(f_{j+1}^{-n} - f_{j-1}^{-n})}{(2\Delta x)}; \quad (3.7)$$

in the downstream region of a shock,

$$\left(\frac{\partial u}{\partial t}\right)_j^n = -\frac{(-3f_j^{-n} + 4f_{j+1}^{-n} - f_{j+2}^{-n})}{(2\Delta x)} - \frac{(f_{j+1}^{+n} - f_{j-1}^{+n})}{(2\Delta x)}. \quad (3.8)$$

Equations (3.7) and (3.8) can be combined together using the expressions for the numerical flux function to form

$$\left(\frac{\partial u}{\partial t}\right)_j^n = -\frac{(h_{j+1/2}^n - h_{j-1/2}^n)}{\Delta x}, \quad (3.9)$$

where

$$h_{j+1/2} = f_{j+1/2L}^+ + f_{j+1/2R}^-, \quad (3.10)$$

and

$$f_{j+1/2L}^+ = \begin{cases} f_j^+ + \frac{1}{2} \Delta f_{j-1/2}^+ & (\text{upstream}), \\ f_j^+ + \frac{1}{2} \Delta f_{j+1/2}^+ & (\text{downstream}); \end{cases} \quad (3.11)$$

$$f_{j+1/2R}^- = \begin{cases} f_{j+1}^- - \frac{1}{2} \Delta f_{j+1/2}^- & (\text{upstream}), \\ f_{j+1}^- - \frac{1}{2} \Delta f_{j+3/2}^- & (\text{downstream}); \end{cases} \quad (3.12)$$

$$\Delta f_{j+1/2}^\pm = f_{j+1}^\pm - f_j^\pm. \quad (3.13)$$

It is clear from Eq. (3.11) that we have to make a choice between $\Delta f_{j-1/2}^+$ and $\Delta f_{j+1/2}^+$ for the upstream and downstream regions. For the shock wave with monotonic u profile, it is easy to convince ourselves that, in general up to second-order accuracy, $\Delta f_{j-1/2}^+$ and $\Delta f_{j+1/2}^+$ have the same algebraic sign, and that in the upstream region, $|\Delta f_{j-1/2}^+|$ is smaller than $|\Delta f_{j+1/2}^+|$, and the opposite is true in the downstream region. Similarly, $\Delta f_{j+1/2}^-$ and $\Delta f_{j+3/2}^-$ also have the same sign, and the absolute value of $\Delta f_{j+1/2}^-$ is smaller than that of $\Delta f_{j+3/2}^-$ in the upstream region, and the opposite is true in the downstream region. Therefore, if we introduce a minmod function, such that $\min\text{mod}(x, y)$ is equal to the argument with the smaller absolute value in the case where x and y have the same algebraic sign, then Eqs. (3.11) and (3.12) can be written as

$$f_{j+1/2L}^+ = f_j^+ + \frac{1}{2} \min\text{mod}(\Delta f_{j-1/2}^+, \Delta f_{j+1/2}^+), \quad (3.14)$$

$$f_{j+1/2R}^- = f_{j+1}^- - \frac{1}{2} \min\text{mod}(\Delta f_{j+1/2}^-, \Delta f_{j+3/2}^-). \quad (3.15)$$

We have remarked that there might exist individual points where the increased entropy condition is not satisfied and the finite-difference solution might possess undesirable oscillations, so we impose, as a choice, when x and y are of different algebraic signs, $\min\text{mod}(x, y) = 0$, and then the finite-difference scheme is reduced to the first-order accuracy.

The Eqs. (3.9), (3.10), (3.14), and (3.15) have been pursued in order to establish the so-called NND scheme. The schemes are second-order accurate (except possibly at some extremum points), and they are dissipative in the

sense that $v_4 < 0$. Therefore, in comparison with other mixed schemes that are first-order accurate in the shock region and second-order accurate elsewhere, the present scheme would be capable of the high resolution of shock waves.

We can prove that the semi-discretized NND scheme possesses the TVD property. In fact, according to the definition of total variation diminishing (TVD) given by Harten, the total variation is

$$TV^n = \sum_j |u_{j+1}^n - u_j^n| = \sum_j |\Delta u_{j+1/2}^n|.$$

Let

$$S_{j+1/2} = \begin{cases} 1, & \text{when } \Delta u_{j+1/2} \geq 0, \\ -1, & \text{when } \Delta u_{j+1/2} < 0. \end{cases}$$

We have

$$\left(\frac{\partial TV}{\partial t}\right)^n = \sum_j S_{j+1/2} \left[\left(\frac{\partial u_{j+1}}{\partial t}\right)^n - \left(\frac{\partial u_j}{\partial t}\right)^n \right]. \quad (3.16)$$

Using (3.10), (3.14), and (3.15), (3.9) can be rewritten as

$$\left(\frac{\partial u_j}{\partial t}\right)^n = -\hat{\alpha}_{j-1/2} \Delta u_{j-1/2}^n + \hat{\beta}_{j+1/2} \Delta u_{j+1/2}^n, \quad (3.17)$$

where

$$\hat{\alpha}_{j-1/2} = \frac{1}{\Delta x} \frac{f_j^{+n} - f_{j-1}^{+n}}{\Delta u_{j-1/2}^n} [1 - \frac{1}{2}(k_2 - k_1)],$$

$$\hat{\beta}_{j+1/2} = -\frac{1}{\Delta x} \frac{f_{j+1}^{-n} - f_j^{-n}}{\Delta u_{j+1/2}^n} [1 - \frac{1}{2}(k_3 - k_4)],$$

and

$$0 \leq k_1 = \min\left(1, \frac{\Delta f_{j+1/2}^{+n}}{\Delta f_{j-1/2}^{+n}}\right) \leq 1,$$

$$0 \leq k_2 = \min\left(\frac{\Delta f_{j-3/2}^{+n}}{\Delta f_{j-1/2}^{+n}}, 1\right) \leq 1,$$

$$0 \leq k_3 = \min\left(1, \frac{\Delta f_{j+3/2}^{-n}}{\Delta f_{j+1/2}^{-n}}\right) \leq 1,$$

$$0 \leq k_4 = \min\left(\frac{\Delta f_{j-1/2}^{-n}}{\Delta f_{j+1/2}^{-n}}, 1\right) \leq 1,$$

Since

$$\frac{f_j^{+n} - f_{j-1}^{+n}}{u_j^n - u_{j-1}^n} \geq 0, \quad \frac{f_{j+1}^{-n} - f_j^{-n}}{u_{j+1}^n - u_j^n} \leq 0,$$

we obtain

$$\hat{\alpha}_{j+1/2} \geq 0, \quad \hat{\beta}_{j+1/2} \geq 0$$

Substituting (3.17) into (3.16), we have

$$\left(\frac{\partial \text{TV}}{\partial t} \right)^n = - \sum_j v_{j+1/2} |\Delta u_{j+1/2}^n|, \quad (3.18)$$

where

$$v_{j+1/2} = \hat{\alpha}_{j+1/2} \left(1 - \frac{S_{j+3/2}}{S_{j+1/2}} \right) + \hat{\beta}_{j+1/2} \left(1 - \frac{S_{j-1/2}}{S_{j+1/2}} \right) \geq 0.$$

It is proven that $(\partial \text{TV}/\partial t)^n \leq 0$; i.e., the semi-discretized NND scheme is total variation diminishing.

B. EXTENSION TO EULER AND NAVIER-STOKES EQUATIONS

It is not difficult to extend formally the scheme for the scalar Eq. (3.1) to one-dimensional Euler equations

$$\frac{\partial U}{\partial t} + \frac{\partial F(U)}{\partial x} = 0. \quad (3.19)$$

Here, U is a vector and F is a function of the vector U . The NND scheme reads, in semi-discretized form,

$$\left(\frac{\partial U}{\partial t} \right)_j^n = -(H_{j+1/2}^n - H_{j-1/2}^n) / \Delta x, \quad (3.20)$$

where

$$H_{j+1/2} = F_{j+1/2L}^+ + F_{j+1/2R}^-, \quad (3.21)$$

$$F_{j+1/2L}^+ = F_j^+ + \frac{1}{2} \min \text{mod}(\Delta F_{j-1/2}^+, \Delta F_{j+1/2}^+), \quad (3.22)$$

$$F_{j+1/2R}^- = F_{j+1}^- - \frac{1}{2} \min \text{mod}(\Delta F_{j+1/2}^-, \Delta F_{j+3/2}^-). \quad (3.23)$$

Now, we study one-dimensional Navier-Stokes equations

$$\frac{\partial U}{\partial t} + \frac{\partial F}{\partial x} = \frac{\partial F_v}{\partial x}, \quad (3.24)$$

where F_v is due to viscous contribution. For the convective part, the NND scheme is used; for the viscous diffusive part, the usual central-difference schemes can be employed.

For three-dimensional flow, the equations can be written as

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial F}{\partial x} + \frac{\partial G}{\partial y} + \frac{\partial H}{\partial z} = \frac{\partial F_v}{\partial x} + \frac{\partial G_v}{\partial y} + \frac{\partial H_v}{\partial z}, \quad (3.25)$$

where the right-hand side is due to viscous contribution. The well-known time-splitting method may be adopted; the problem is reduced to three 1D problems:

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial F}{\partial x} = \frac{\partial F_v}{\partial x}, \quad (3.26)$$

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial G}{\partial y} = \frac{\partial G_v}{\partial y}, \quad (3.27)$$

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial H}{\partial z} = \frac{\partial H_v}{\partial z}. \quad (3.28)$$

For every 1D problem, the NND scheme can be used. Define L_x, L_y, L_z for the difference expressions of Eqs. (3.26), (3.27), and (3.28). Then, we have

$$\mathbf{U}^{n+1} = L(\Delta t) \mathbf{U}^n.$$

$L(\Delta t)$ has one of the following forms:

$$L(\Delta t) = L_x(\Delta t/2) L_y(\Delta t/2) L_z(\Delta t) L_y(\Delta t/2) L_x(\Delta t/2)$$

or

$$L(\Delta t) = L_y(\Delta t/2) L_z(\Delta t/2) L_x(\Delta t) L_z(\Delta t/2) L_y(\Delta t/2)$$

$$L(\Delta t) = L_z(\Delta t/2) L_x(\Delta t/2) L_y(\Delta t) L_x(\Delta t/2) L_z(\Delta t/2)$$

For Euler equations, we have the Δt in the preceding expression chosen as

$$\Delta t = \min(\Delta t_x, \Delta t_y, \Delta t_z),$$

where

$$\Delta t_x = CFL_{\max} \Delta x / |\lambda_A|_{\max},$$

$$\Delta t_y = CFL_{\max} \Delta y / |\lambda_B|_{\max},$$

$$\Delta t_z = CFL_{\max} \Delta z / |\lambda_C|_{\max},$$

$|\lambda_A|_{\max}$ is the maximum absolute eigenvalue of the matrix $A = \partial F / \partial \mathbf{U}$, $|\lambda_B|_{\max}$ is the corresponding eigenvalue of the matrix $B = \partial G / \partial \mathbf{U}$, and $|\lambda_C|_{\max}$ is the corresponding eigenvalue of the matrix $C = \partial H / \partial \mathbf{U}$. Similar expressions can be given for Navier–Stokes equations.

IV. Explicit NND Schemes

Based on the preceding semi-discrete NND scheme, the numerical solution of the scalar Eq. (3.1) can be computed by the following five different explicit schemes discretized in time and space.

A. NND-1 SCHEME (see Zhang, 1988)

For this case, $(\partial u / \partial t)_j^n$ in (3.9) is discretized using the Euler first-order difference, i.e., $(\partial u / \partial t)_j^n = (u_j^{n+1} - u_j^n) / \Delta t$. Then, from (3.9), we have

$$u_j^{n+1} = u_j^n - \frac{\Delta t}{\Delta x} (h_{j+1/2}^n - h_{j-1/2}^n). \quad (4.1)$$

This scheme is first order in time and second order in space. It possesses the TVD property. The maximum allowable Courant number is $2/3$. The scheme is referred to as NND-1.

Through a rearrangement for $h_{j+1/2}^n$, we can rewrite it as follows:

$$\begin{aligned} h_{j+1/2} &= \frac{1}{2} [f_j + f_{j+1} - (\alpha_{j+1/2}^+ - \alpha_{j+1/2}^-) \Delta u_{j+1/2}] \\ &\quad + \frac{1}{2} \min \text{mod}(\Delta t_{j-1/2}^+, \beta \Delta f_{j+1/2}^+) \\ &\quad - \frac{1}{2} \min \text{mod}(\Delta f_{j+1/2}^-, \beta \Delta f_{j+3/2}^-), \end{aligned} \quad (4.2)$$

with $\beta = 1$. Osher and Chakravarthy (1984) have obtained a TVD scheme using the flux limiter concept by simply modifying the second-order upwind scheme. Our numerical flux expression (4.2) is formally just the same as theirs except for the parameter β . However, we notice that at $\beta = 2$ (see Yee, 1985) the general effect of this limiter is to reduce the scheme to first-order spatial accuracy in regions of steep gradients (see Thompson and Matus, 1989). However, NND-1 has second-order accuracy in regions of steep gradients.

For the Euler equation (3.19), the NND-1 scheme can be expressed as

$$U_j^{n+1} = U_j^n - \frac{\Delta t}{\Delta x} (H_{j+1/2}^n - H_{j-1/2}^n), \quad (4.3)$$

where

$$\begin{aligned} H_{j+1/2} &= F_j^+ + \frac{1}{2} \min \text{mod}(\Delta F_{j+1/2}^+, \Delta F_{j-1/2}^+) \\ &\quad + F_{j+1}^- - \frac{1}{2} \min \text{mod}(\Delta F_{j+3/2}^-, \Delta F_{j+1/2}^-). \end{aligned}$$

B. NND-2 SCHEME (see Zhang, 1989b)

Using the Taylor expression and (3.1), we obtain

$$u_j^{n+1} = u_j^n + \left(\frac{\partial u}{\partial t}\right)_j \Delta t + \frac{1}{2} \Delta t^2 \frac{\partial}{\partial x} \left(\alpha \frac{\partial f}{\partial x}\right) + \dots, \quad (4.4)$$

where $f = \alpha u$. When $\alpha = \alpha^+ > 0$, we have $f = f^+ = \alpha^+ u$ and, when $\alpha = \alpha^- < 0$, then $f = f^- = \alpha^- u$. Since $\partial/\partial x(\alpha \partial f/\partial x)$ may be expressed as

$$\begin{aligned} \frac{\partial}{\partial x} \left(\alpha \frac{\partial f}{\partial x}\right) &= \frac{1}{\Delta x^2} \left[\text{minmod}(\alpha_{j+1/2}^+ \Delta f_{j+1/2}^+, \alpha_{j-1/2}^+ \Delta f_{j-1/2}^+) \right. \\ &\quad \left. - \text{minmod}(\alpha_{j-1/2}^+ \Delta f_{j-1/2}^+, \alpha_{j-3/2}^+ \Delta f_{j-3/2}^+) \right] \\ &\quad + \frac{1}{\Delta x^2} \left[\text{minmod}(\alpha_{j+3/2}^- \Delta f_{j+3/2}^-, \alpha_{j+1/2}^- \Delta f_{j+1/2}^-) \right. \\ &\quad \left. - \text{minmod}(\alpha_{j+1/2}^- \Delta f_{j+1/2}^-, \alpha_{j-1/2}^- \Delta f_{j-1/2}^-) \right]. \end{aligned} \quad (4.5)$$

The numerical solution of (3.1) may be expressed as

$$u_j^{n+1} = u_j^n - \frac{\Delta t}{\Delta x} (\tilde{h}_{j+1/2}^n - \tilde{h}_{j-1/2}^n), \quad (4.6)$$

where

$$\begin{aligned} \tilde{h}_{j+1/2} &= f_j^+ + \frac{1}{2} \text{minmod} \left[\left(1 - \alpha_{j+1/2}^+ \frac{\Delta t}{\Delta x}\right) \Delta f_{j+1/2}^+, \right. \\ &\quad \left. \left(1 - \alpha_{j-1/2}^+ \frac{\Delta t}{\Delta x}\right) \Delta f_{j-1/2}^+ \right] \\ &\quad + f_{j+1}^- - \frac{1}{2} \text{minmod} \left[\left(1 + \alpha_{j+3/2}^- \frac{\Delta t}{\Delta x}\right) \Delta f_{j+3/2}^-, \right. \\ &\quad \left. \left(1 + \alpha_{j+1/2}^- \frac{\Delta t}{\Delta x}\right) \Delta f_{j+1/2}^- \right]. \end{aligned}$$

This TVD scheme is second order in time and space. The maximum allowable Courant number is 1. We refer to this scheme as NND-2.

Through some rearrangements for $\tilde{h}_{j+1/2}$, we can prove that NND-2 is just the Harten TVD scheme (see Harten, 1983).

For the Euler equation (3.19), NND-2 can be written as

$$\mathbf{U}_j^{n+1} = \mathbf{U}_j^n - \frac{\Delta t}{\Delta x} (\tilde{H}_{j+1/2}^n - \tilde{H}_{j-1/2}^n), \quad (4.7)$$

where

$$\begin{aligned} \tilde{H}_{j+1/2} = & F_j^+ + \frac{1}{2} \min \text{mod} \left[\left(I - A_{j+1/2}^+ \frac{\Delta t}{\Delta x} \right) \Delta F_{j+1/2}^+, \right. \\ & \left. \left(I - A_{j-1/2}^+ \frac{\Delta t}{\Delta x} \right) \Delta F_{j-1/2}^+ \right] \\ & + F_{j+1}^- - \frac{1}{2} \min \text{mod} \left[\left(I + A_{j+3/2}^- \frac{\Delta t}{\Delta x} \right) \Delta F_{j+3/2}^-, \right. \\ & \left. \left(I + A_{j+1/2}^- \frac{\Delta t}{\Delta x} \right) \Delta F_{j+1/2}^- \right]. \end{aligned}$$

C. NND-3 SCHEME (see Zhang, 1989b)

This is a two-step predictor–corrector scheme. For a scalar hyperbolic equation (3.1), the scheme is

$$\begin{cases} u_j^{\overline{n+1}} = u_j^n - \frac{1}{2} \frac{\Delta t}{\Delta x} (h_{j+1/2}^n - h_{j-1/2}^n), \\ u_j^{n+1} = u_j^n - \frac{\Delta t}{\Delta x} (h_{j+1/2}^{\overline{n+1}} - h_{j-1/2}^{\overline{n+1}}), \end{cases} \quad (4.8)$$

where $h_{j+1/2}$ is given by (3.10), (3.14), and (3.15). This TVD scheme is second order in time and space. The maximum allowable Courant number is 1. We refer to this as NND-3 scheme.

For the one-dimensional Euler equation (3.19), NND-3 scheme can be expressed as

$$\begin{aligned} \mathbf{U}_j^{\overline{n+1}} &= \mathbf{U}_j^n - \frac{\Delta t}{2 \Delta x} (H_{j+1/2}^n - H_{j-1/2}^n), \\ \mathbf{U}_j^{n+1} &= \mathbf{U}_j^n - \frac{\Delta t}{\Delta x} (H_{j+1/2}^{\overline{n+1}} - H_{j-1/2}^{\overline{n+1}}), \end{aligned}$$

where $H_{j+1/2}$ is given by (3.21), (3.22), and (3.23).

D. NND-4 SCHEME (see Zhang, 1989b)

This is another two-step predictor–corrector scheme. For Eq. (3.1), the scheme is

$$\begin{cases} u_j^{\overline{n+1}} = u_j^n - \frac{\Delta t}{\Delta x} (h_{j+1/2}^n - h_{j-1/2}^n), \\ u_j^{n+1} = \frac{1}{2} \left[u_j^n + u_j^{\overline{n+1}} - \frac{\Delta t}{\Delta x} (h_{j+1/2}^{\overline{n+1}} - h_{j-1/2}^{\overline{n+1}}) \right], \end{cases} \quad (4.9)$$

This scheme is second order in time and space, and is a TVD scheme. The maximum allowable Courant number is 1.

For simplicity, the predictor step of (4.9) can be also taken as

$$u_j^{\overline{n+1}} = u_j^n - \frac{\Delta t}{\Delta x} (\Delta f_{j-1/2}^{+n} + \Delta f_{j+1/2}^{-n}),$$

i.e., $\minmod(a, b) = 0$ is used in $h_{j+1/2}^n$. Then, the scheme, which is still second order in time and space, can be expressed as

$$\begin{cases} u_j^{\overline{n+1}} = u_j^n - \frac{\Delta t}{\Delta x} (\Delta f_{j-1/2}^{+n} + \Delta f_{j+1/2}^{-n}), \\ u_j^{n+1} = u_j^n - \frac{\Delta t}{\Delta x} (\tilde{h}_{j+1/2} - \tilde{h}_{j-1/2}), \end{cases} \quad (4.10)$$

where

$$\begin{aligned} \tilde{h}_{j+1/2} &= \frac{1}{2}(f_j^{+n} + f_j^{\overline{n+1}}) + \frac{1}{2} \minmod(\Delta f_{j+1/2}^{+\overline{n+1}}, \Delta f_{j-1/2}^{+\overline{n+1}}) \\ &\quad + \frac{1}{2}(f_{j+1}^{-n} + f_{j+1}^{\overline{n+1}}) - \frac{1}{2} \minmod(\Delta f_{j+3/2}^{-\overline{n+1}}, \Delta f_{j+1/2}^{-\overline{n+1}}). \end{aligned} \quad (4.11)$$

This is also a TVD scheme. The maximum allowable Courant number is 1. We refer to (4.9) and (4.10) as NND-4(a) and NND-4(b).

Substituting (4.11) into (4.10), we obtain

$$\begin{aligned} u_j^{\overline{n+1}} &= u_j^n - \frac{\Delta t}{\Delta x} (\Delta f_{j-1/2}^{+n} + \Delta f_{j+1/2}^{-n}), \\ u_j^{n+1} &= \frac{1}{2} \left[u_j^n + u_j^{\overline{n+1}} - \frac{\Delta t}{\Delta x} (\Delta f_{j+1/2}^{+\overline{n+1}} + \Delta f_{j-1/2}^{-\overline{n+1}}) \right] \\ &\quad + \frac{1}{2} \frac{\Delta t}{\Delta x} [\Delta f_{j+1/2}^{+\overline{n+1}} - \minmod(\Delta f_{j-1/2}^{+\overline{n+1}}, \Delta f_{j-1/2}^{-\overline{n+1}})] \end{aligned}$$

$$\begin{aligned}
& -\frac{1}{2} \frac{\Delta t}{\Delta x} [\Delta f_{j-1/2}^{+\bar{n}+1} - \min\text{mod}(\Delta f_{j-1/2}^{+\bar{n}+1}, \Delta f_{j-3/2}^{+\bar{n}+1})] \\
& -\frac{1}{2} \frac{\Delta t}{\Delta x} [\Delta f_{j+1/2}^{-\bar{n}+1} - \min\text{mod}(\Delta f_{j+1/2}^{-\bar{n}+1}, \Delta f_{j+3/2}^{-\bar{n}+1})] \\
& +\frac{1}{2} \frac{\Delta t}{\Delta x} [\Delta f_{j+1/2}^{-\bar{n}+1} - \min\text{mod}(\Delta f_{j+1/2}^{-\bar{n}+1}, \Delta f_{j-1/2}^{-\bar{n}+1})]. \quad (4.12)
\end{aligned}$$

It is very clear that NND-4(b) is the extension of MacCormack's two-step predictor-corrector scheme. Because some third-order terms shown in (4.12) are added to MacCormack's corrector step, NND-4(b) is a TVD scheme.

For one-dimensional Euler equations, NND-4 can be written as

$$\bar{\mathbf{U}}_j^{\bar{n}+1} = \mathbf{U}_j^n - \frac{\Delta t}{\Delta x} (\Delta F_{j-1/2}^{+\bar{n}} + \Delta F_{n+1/2}^{-\bar{n}}),$$

$$\mathbf{U}_j^{n+1} = \mathbf{U}_j^n - \frac{\Delta t}{\Delta x} (\tilde{H}_{j+1/2} - \tilde{H}_{j-1/2}),$$

where

$$\begin{aligned}
\tilde{H}_{j+1/2} &= \frac{1}{2}(F_j^{+\bar{n}} + F_j^{+\bar{n}+1}) + \frac{1}{2} \min\text{mod}(\Delta F_{j+1/2}^{+\bar{n}+1}, \Delta F_{j-1/2}^{+\bar{n}+1}) \\
&+ \frac{1}{2}(F_{j+1}^{-\bar{n}} + F_{j+1}^{-\bar{n}+1}) - \frac{1}{2} \min\text{mod}(\Delta F_{j+3/2}^{-\bar{n}+1}, \Delta F_{j+1/2}^{-\bar{n}+1}).
\end{aligned}$$

E. NND-5 SCHEME (see Zhang, 1989b)

According to the Runge-Kutta method, the numerical solution of (3.1) can be expressed as

$$u_j^{(1)} = u_j^n - \alpha_1 \frac{\Delta t}{\Delta x} (h_{j+1/2}^n - h_{j-1/2}^n),$$

$$u_j^{(2)} = u_j^n - \alpha_2 \frac{\Delta t}{\Delta x} (h_{j+1/2}^{(1)} - h_{j-1/2}^{(1)}),$$

$$u_j^{n+1} = u_j^n - \alpha_3 \frac{\Delta t}{\Delta x} (h_{j+1/2}^{(2)} - h_{j-1/2}^{(2)}).$$

In this paper, we assume $\alpha_1 = 1/6$, $\alpha_2 = 1/2$, $\alpha_3 = 1$. The scheme is second order in time and space. The maximum allowable Courant number is 2.

If we substitute $\mathbf{U}, F$ for u, f , respectively, the NND-5 scheme solving one-dimensional Euler equations is obtained.

V. Implicit NND Scheme (see Zhang, 1989a)

The numerical results show that the preceding explicit NND schemes possess a good convergence accuracy and a high resolution for capturing shock waves, vortices, and shear layers. However, the computational time is long for explicit calculations with fine mesh, such as the case for computing the turbulent flows. Therefore, it is necessary to develop an implicit NND scheme for the unconditional stability.

Some implicit TVD schemes have been presented in the references. Because the implicit difference of these schemes depends respectively on 5, 9, 13 grid points for one- two- and three-dimensional flows in general, the amount of numerical work for inverting diagonal matrices with blocks are large. It is necessary to develop implicit schemes that depend on a smaller number of grid points.

Based on the preceding considerations, a new implicit NND scheme is developed. The details will be described in what follows.

In fact, for the one-dimensional Euler equation (3.19), we have

$$\left(\frac{\partial \mathbf{U}}{\partial t}\right)_j^n = -\frac{1}{\Delta x}(H_{j+1/2}^n - H_{j-1/2}^n), \quad (5.1)$$

$$\left(\frac{\partial \mathbf{U}}{\partial t}\right)_j^{n+1} = -\frac{1}{\Delta x}(H_{j+1/2}^{n+1} - H_{j-1/2}^{n+1}), \quad (5.2)$$

from (3.20). According to the Crank-Nicolson method, $\mathbf{U}_j^{n+1}$ can be expressed as

$$\mathbf{U}_j^{n+1} = \mathbf{U}_j^n + \Delta t \left[(1 - \theta) \left(\frac{\partial \mathbf{U}}{\partial t}\right)_j^n + \theta \left(\frac{\partial \mathbf{U}}{\partial t}\right)_j^{n+1} \right]. \quad (5.3)$$

The expression (5.3) is of second-order accuracy in time for $\theta = \frac{1}{2}$ and is first-order for $\theta = 1$. Substituting (5.1) and (5.2) into this expression, the following implicit scheme can be obtained:

$$\mathbf{U}_j^{n+1} = \mathbf{U}_j^n - \frac{\Delta t}{\Delta x}(\tilde{H}_{j+1/2} - \tilde{H}_{j-1/2}), \quad (5.4)$$

where

$$\begin{aligned} \tilde{H}_{j+1/2} - \tilde{H}_{j-1/2} = & [(1 - \theta)H_{j+1/2}^n + \theta H_{j+1/2}^{n+1}] \\ & - [(1 - \theta)H_{j-1/2}^n + \theta H_{j-1/2}^{n+1}]. \end{aligned} \quad (5.5)$$

Define

$$\begin{aligned} k_1 &= \frac{1}{2} \min\left(\frac{\Delta F_{j+1/2}^+}{\Delta F_{j-1/2}^+}, 1\right), \\ k_2 &= \frac{1}{2} \min\left(\frac{\Delta F_{j+3/2}^-}{\Delta F_{j+1/2}^-}, 1\right), \\ k_3 &= \frac{1}{2} \min\left(1, \frac{\Delta F_{j-3/2}^+}{\Delta F_{j-1/2}^+}\right), \\ k_4 &= \frac{1}{2} \min\left(1, \frac{\Delta F_{j-1/2}^-}{\Delta F_{j+1/2}^-}\right). \end{aligned}$$

The main idea in establishing our implicit scheme is to use the limiters k_s at time step n ; we have, in general,

$$k_j^{n+1} \Delta F_{j+1/2}^{\pm n+1} = k_j^n \Delta F_{j+1/2}^{\pm n+1} + O(\Delta t \Delta x).$$

Then, Eq. (5.5) may be written as

$$\begin{aligned} \tilde{H}_{j+1/2} - \tilde{H}_{j-1/2} &= H_{j+1/2}^n - H_{j-1/2}^n + \theta(1 + k_1^n - k_3^n)(\delta F_j^{+n+1} - \delta F_{j-1}^{+n+1}) \\ &\quad + \theta(1 - k_2^n + k_4^n)(\delta F_{j+1}^{-n+1} - \delta F_j^{-n+1}), \end{aligned}$$

where

$$\delta F_j^{\pm n+1} = F_j^{\pm n+1} - F_j^{\pm n}.$$

Since

$$\delta F_j^{+n+1} = A_j^{+n} \delta U_j^{n+1} + O(\Delta t^2),$$

$$\delta F_j^{-n+1} = A_j^{-n} \delta U_j^{n+1} + O(\Delta t^2),$$

where

$$\delta U_j^{n+1} = U_j^{n+1} - U_j^n, \quad A^+ + A^- = \frac{\partial F}{\partial U} = A.$$

Equation (5.4) can be estimated by the equation

$$\hat{A}_{j-1}^n \delta U_{j-1}^{n+1} + \hat{B}_j^n \delta U_j^{n+1} + \hat{C}_{j+1}^n \delta U_{j+1}^{n+1} = -\frac{\Delta t}{\Delta x} (H_{j+1/2}^n - H_{j-1/2}^n), \quad (5.6)$$

where

$$\hat{A}_{j-1}^n = -\theta \frac{\Delta t}{\Delta x} (1 + k_1^n - k_3^n) A_{j-1}^{+n},$$

$$\hat{B}_j^n = I + \theta \frac{\Delta t}{\Delta x} (1 + k_1^n - k_3^n) A_j^{+n} - \theta \frac{\Delta t}{\Delta x} (1 - k_2^n + k_4^n) A_j^{-n},$$

$$\hat{C}_{j+1}^n = \theta \frac{\Delta t}{\Delta x} (1 - k_2^n + k_4^n) A_{j+1}^{-n}.$$

Equation (5.6) is the goal we have pursued in order to establish the so-called implicit NND scheme. For $\theta = 1$, the scheme is of first-order accuracy in time and second-order in space. We can prove that this scheme for $\theta = 1$ is a TVD scheme, and is unconditionally stable.

VI. Applications to Solutions of Euler and Navier-Stokes Equations

To test the effectiveness of NND schemes, in addition to model problems, we calculated two- and three-dimensional flows with Euler and Navier-Stokes equations. The computational results are given in this section.

A. REGULAR REFLECTION OF SHOCK WAVE (see Zhang, 1989b)

1. Governing Equations and Boundary Conditions

A popular problem for checking out shock-capturing properties is the regular reflection of an oblique shock wave by a flat plate. Figure 3 shows the computational region. The Euler equations of gas dynamics with $\gamma = 1.4$ in the Cartesian (x, y) plane can be written in the conservative form as

$$\frac{\partial E}{\partial x} + \frac{\partial F}{\partial y} = 0, \quad (6.1)$$

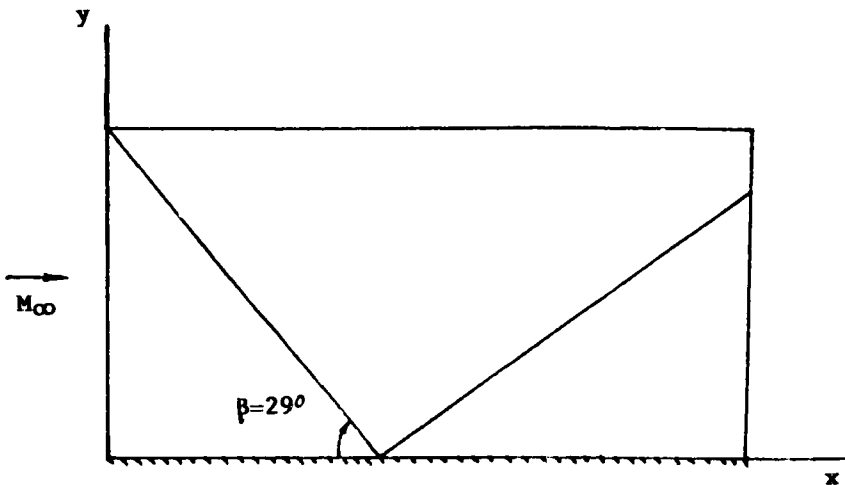


FIG. 3. Regular reflection of an oblique shock wave.

where

$$E = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ (e + p)u \end{bmatrix}, \quad F = \begin{bmatrix} \rho v \\ \rho uv \\ \rho v^2 + p \\ (e + p)v \end{bmatrix}.$$

In the preceding equations, u and v are the velocity components in the x and y direction, respectively, and

$$e = \frac{p}{\gamma - 1} + \frac{1}{2} \rho(u^2 + v^2).$$

We specify a uniform $M_\infty = 2.9$ flow at the left boundary. At the top boundary, we specify the conditions behind a shock, where the angle β made by the incident shock wave and flat plate is 29° . A flow tangency condition is specified at the wall, and all variables are extrapolated at right boundary.

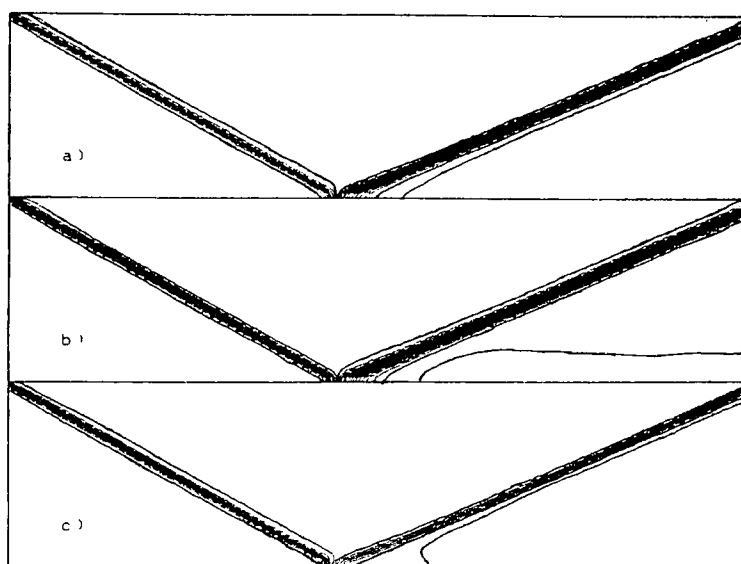
2. Difference Equation

Equation (6.1) is a hyperbolic system for this case, so we can solve it using the space-marching algorithm along the positive direction of x -axis. In fact, since E and F are functions of U where $U = [\rho, \rho u, \rho v, e]^T$, F is a function of E . Substituting E and F here for U and F , respectively, in Section V gives the NND-1-5 schemes.

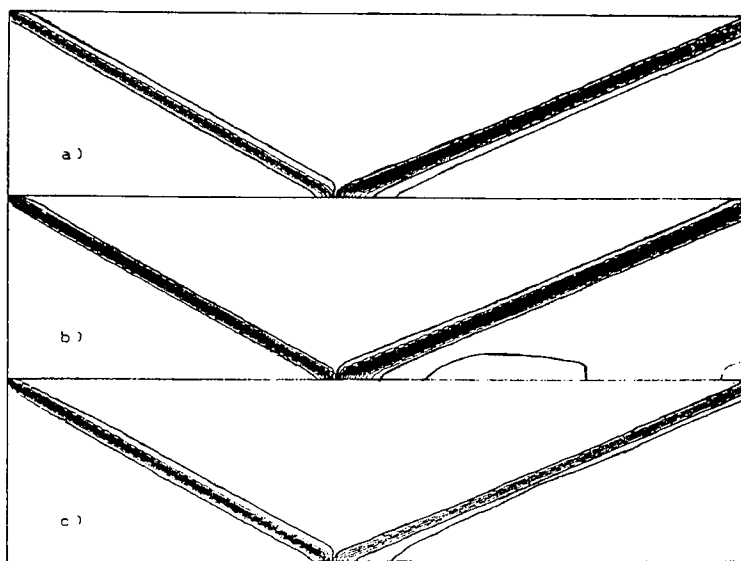
The space-marching computation is started with the left boundary. The grid used is $\Delta x = 1/100$ and $\Delta y = 1/50$.

3. Computational Results (see Zhang, 1989b)

Figure 4 shows the pressure, the density, and the Mach number contours evaluated with NND-1-5. The pressure coefficients evaluated at $y = 0.5$ are shown in Fig. 5. In Figs. 6 and 7, the pressure, density, and Mach number contours and pressure coefficients calculated using first-order upwind scheme (UP-1), second-order upwind scheme (UP-2) and two-step MacCormack's explicit scheme (MC) are shown. It is very clear that the spurious oscillations near or in the shock region can be totally suppressed with every one of the NND schemes 1-5. Therefore, the NND schemes 1-5 possess the distinguished capability of capturing shocks.

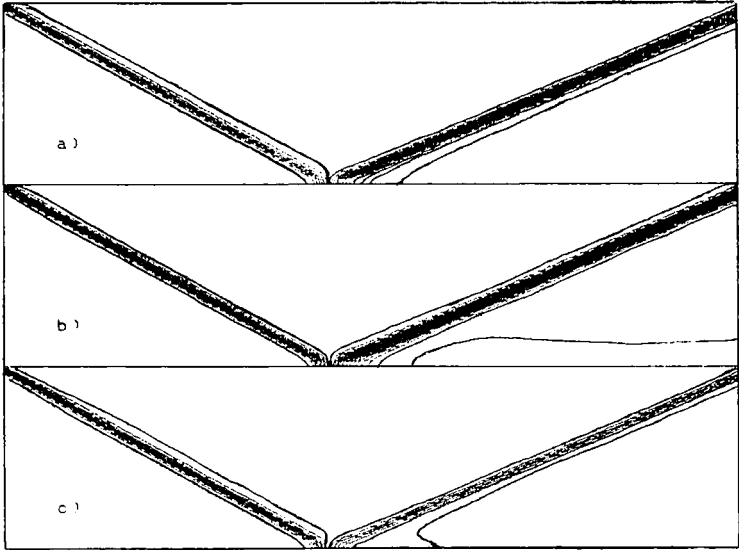


(1) *NND-1*

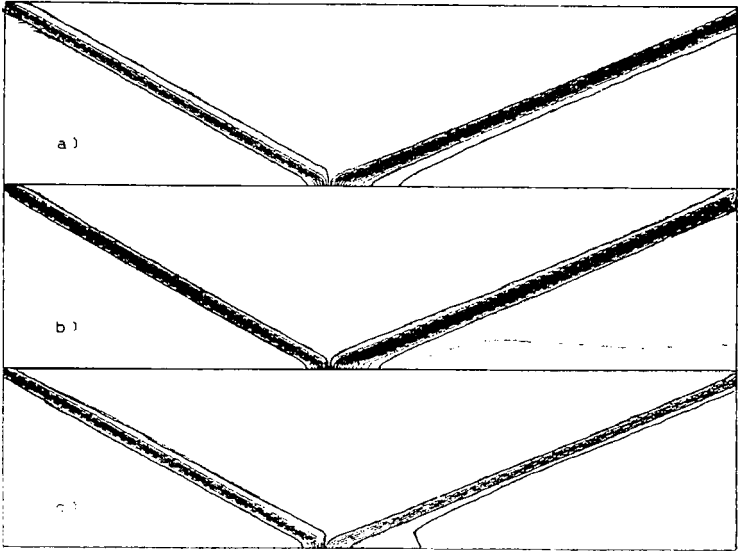


(2) *NND-2*

FIG. 4 The pressure (a), the density (b), and Mach number contours for regular reflection of an oblique shock wave.



(3) NND-3



(4) NND-4A

FIG. 4—continued.

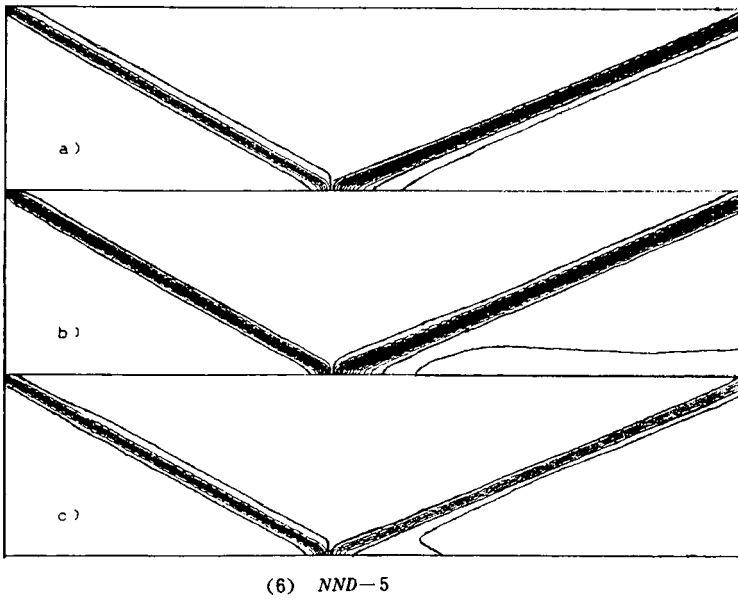
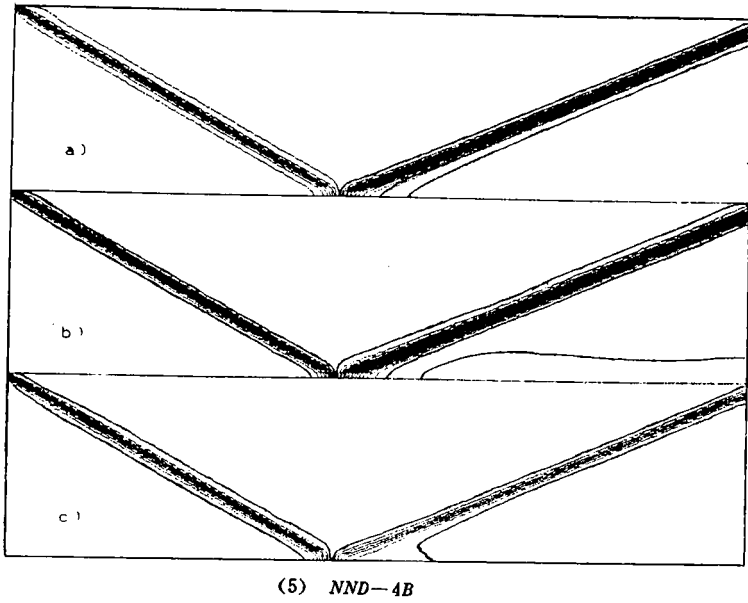
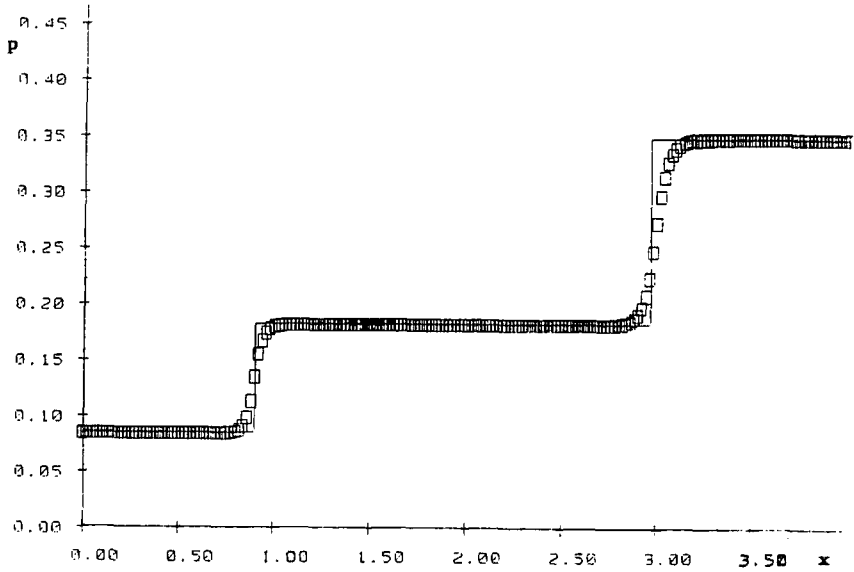
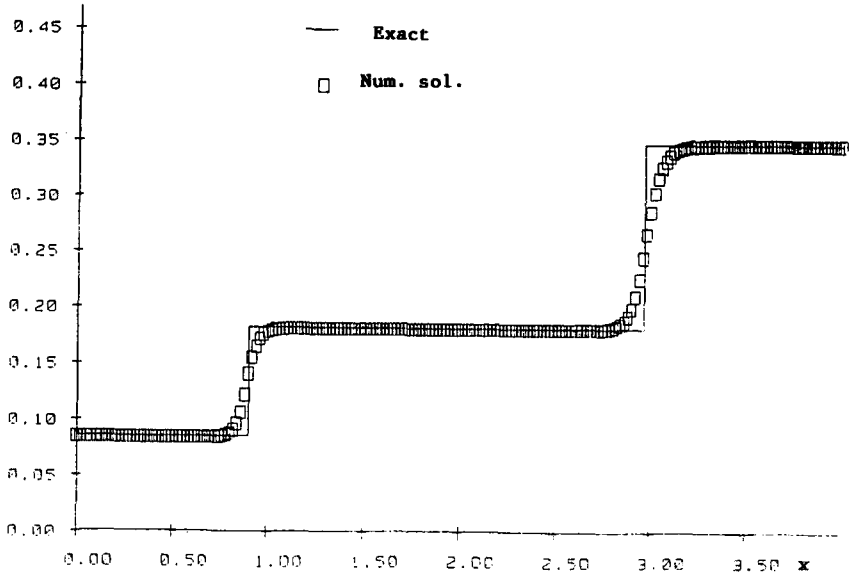


FIG. 4—continued.

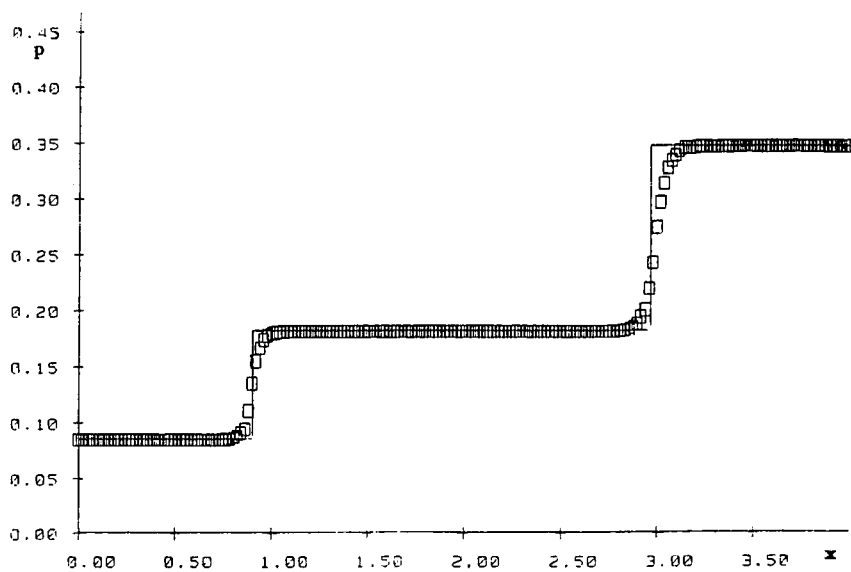


(1) NND-1

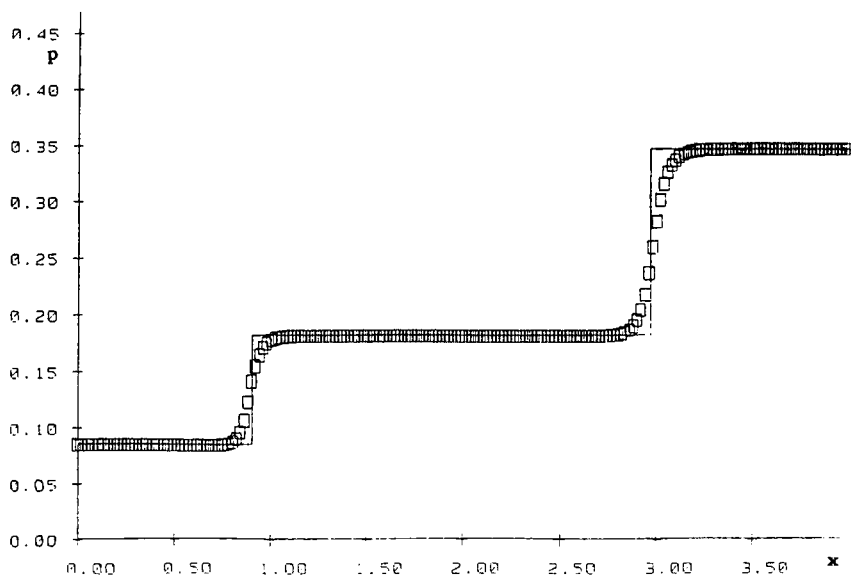


(2) NND-2

FIG. 5 Pressure distributions for regular reflection at $y = 0.5$.

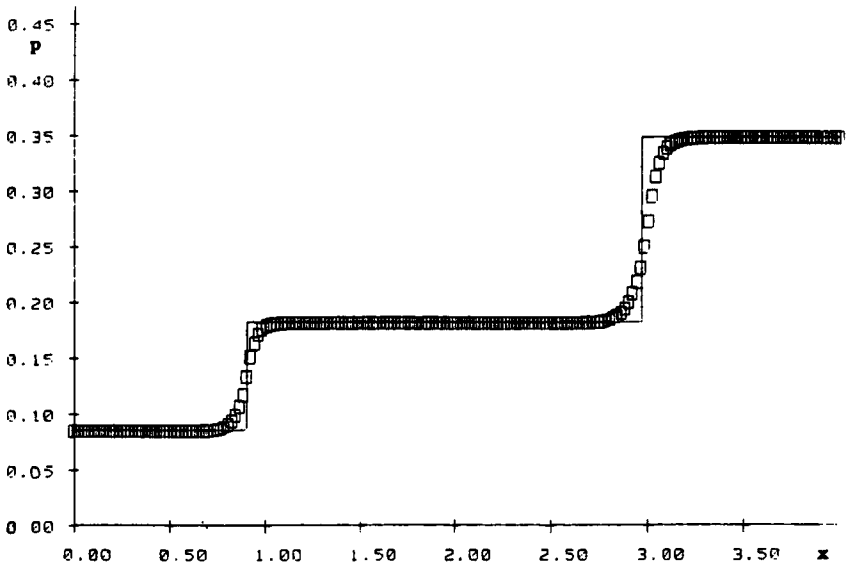


(3)NND-3

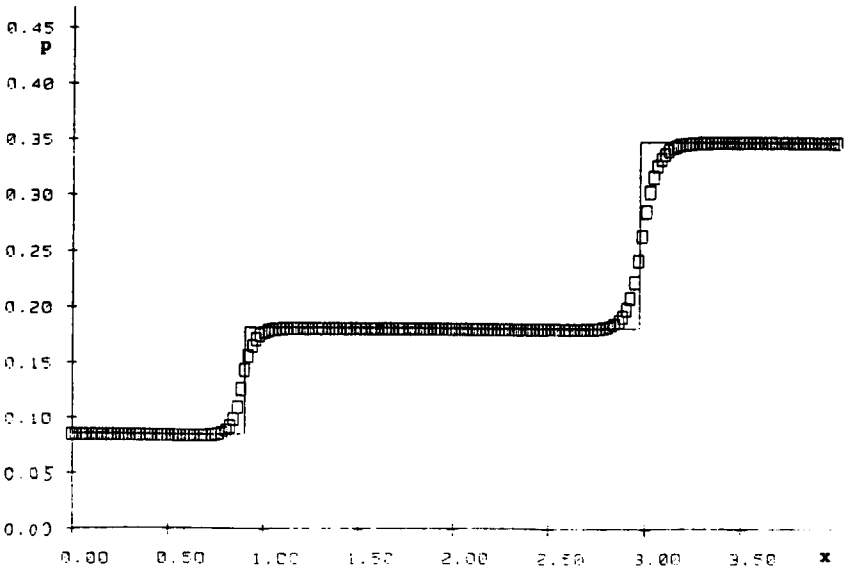


(4)NND-4A

FIG. 5—continued.

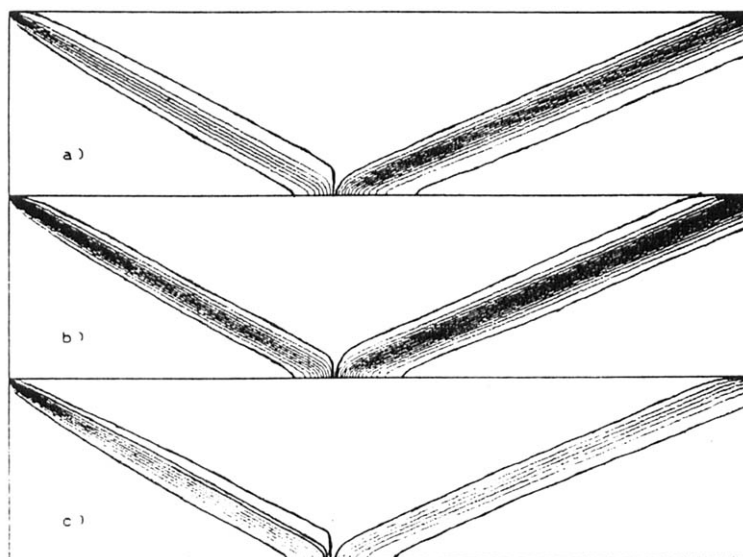


(5)NND-4B

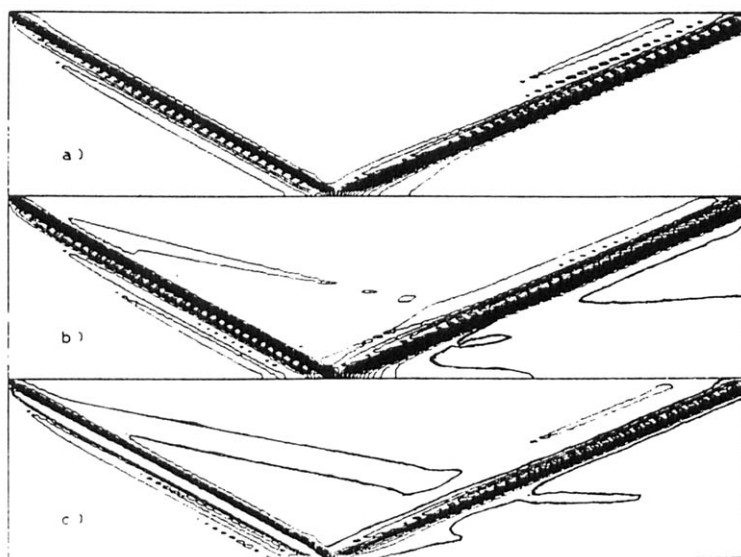


(6)NND-5

FIG. 5—continued.

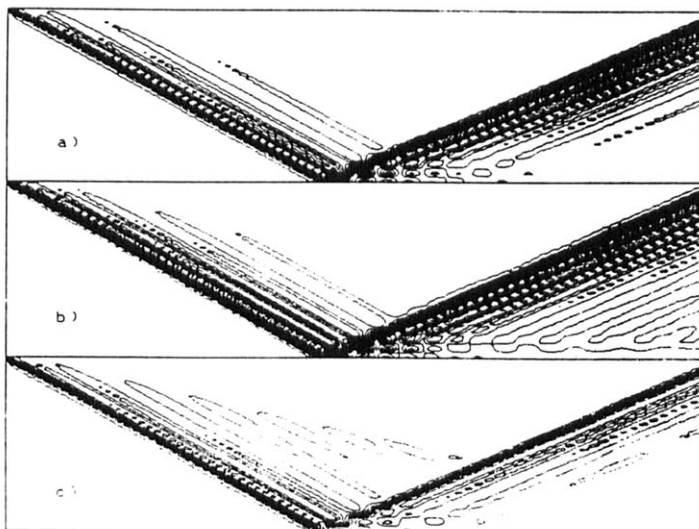


(1) *UP-1*



(2) *UP-2*

FIG. 6 The Pressure (a) density (b) and Mach number (c) contours calculated by UP-1, UP-2 and MC schemes.



(3) MC

FIG. 6—continued.

In Table 6.1, the numerical work required per marching step and the maximum allowable marching step are given for schemes used here. Both indexes show that the amount of numerical work is much reduced for the NND-1, NND-4, and NND-5 schemes.

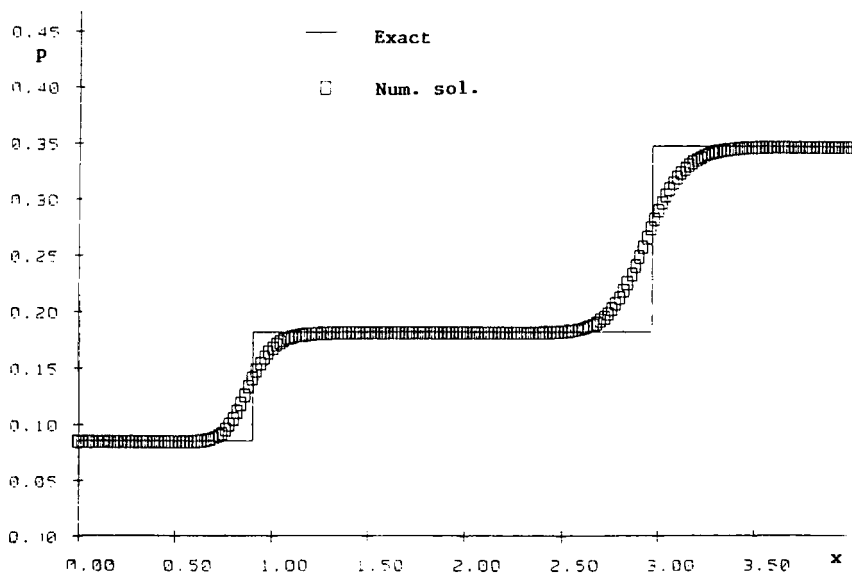
Using the NND-1 and NND-4 schemes, the reflection of shock wave between two flat plates is also simulated. Figure 8(1)–(4) show the pressure, the density, Mach number contours, and pressure profiles at $y = 0.25, 0.50$, and 0.75 . Figure 8(5)–(8) show the results calculated using the first-order and second-order upwind schemes.

TABLE 1

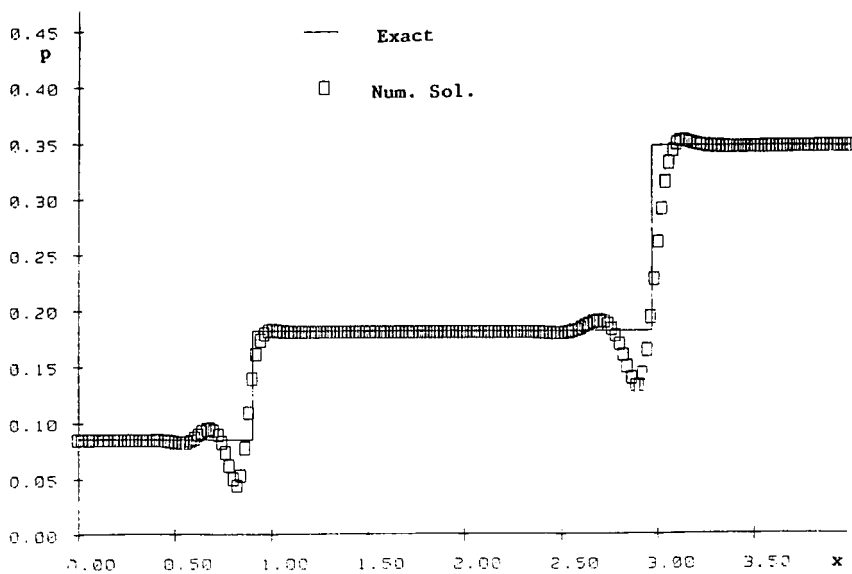
Scheme NND	Numerical work per marching step*	Allowable maximum marching step*
–1	1	1
–2	4.803	1.5
–3	1.882	1.5
–4A	1.453	1.5
–4B	1.483	1.5
5	2.85	3.0

* 1. The data are relative to one of NND-1 scheme.

2. Computer VAX II/780 is used.

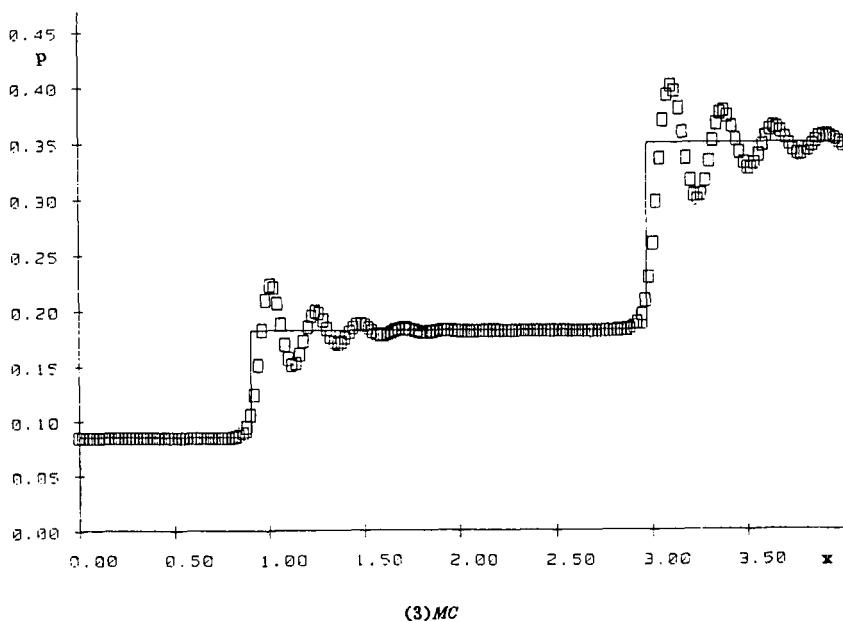


(1)UP-1



(2)UP-2

FIG. 7 The pressure distributions calculated by UP-1, UP-2 and MC schemes.

(3)MC
FIG. 7—continued.

B. SUPERSONIC VISCOUS FLOW AROUND A CIRCULAR CYLINDER (see Zhang, 1989b)

1. The Governing Equations and Boundary Conditions

The conservation laws for the two-dimensional Navier-Stokes equations can be written in the form

$$\frac{\partial u}{\partial t} + \frac{\partial F}{\partial x} + \frac{\partial G}{\partial y} = \frac{\partial F_v}{\partial x} + \frac{\partial G_v}{\partial y}, \quad (6.2)$$

where

$$U = (\rho, \rho u, \rho v, e)^T,$$

$$F = (\rho u, \rho u^2 + p, \rho uv, eu + pu)^T,$$

$$G = (\rho v, \rho vu, \rho v^2 + p, (e + p)v)^T,$$

$$F_v = (0, \tau_{xx}, \tau_{xy}, f)^T,$$

$$G_v = (0, \tau_{xy}, \tau_{yy}, g)^T,$$

and

$$e = \frac{p}{\gamma - 1} + \frac{\rho}{2}(u^2 + v^2).$$

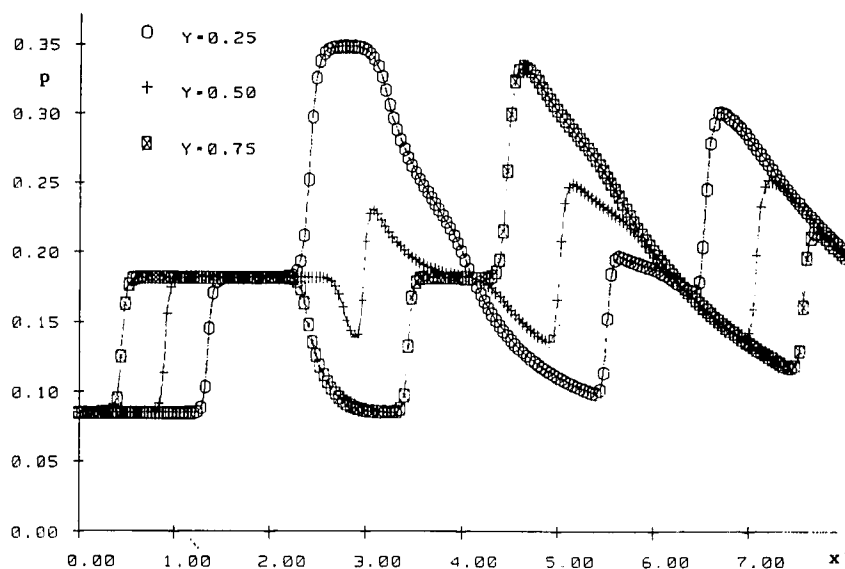


(1) *NND-1*

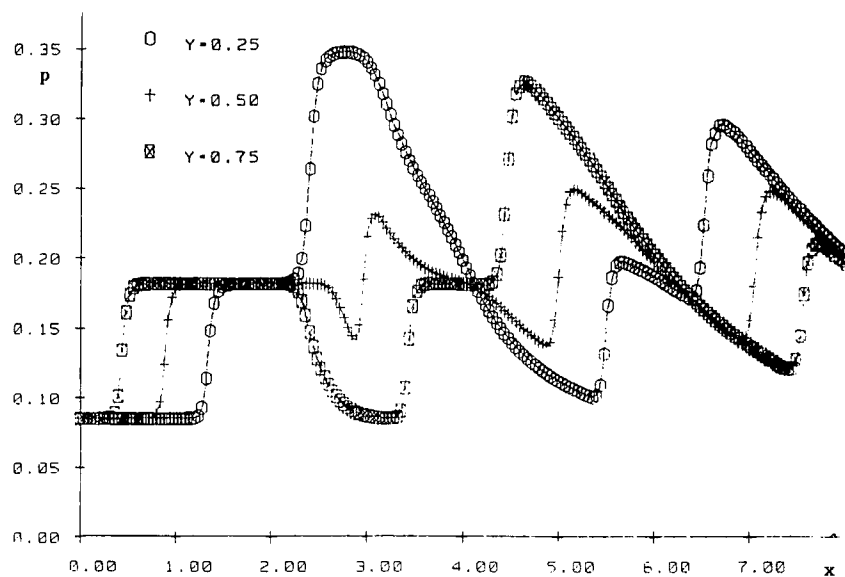


(2) *NND-4A*

FIG. 8 (1)–(4) The pressure (a), density (b), Mach number (c) contours and pressure distributions calculated by NND-1 and NND-4A.



(3) NND-1

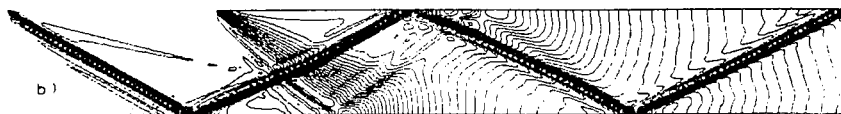
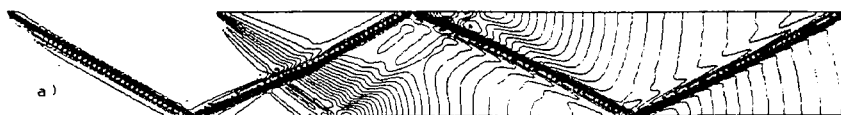


(4) NND-4A

FIG. 8 (1)-(4)—continued.

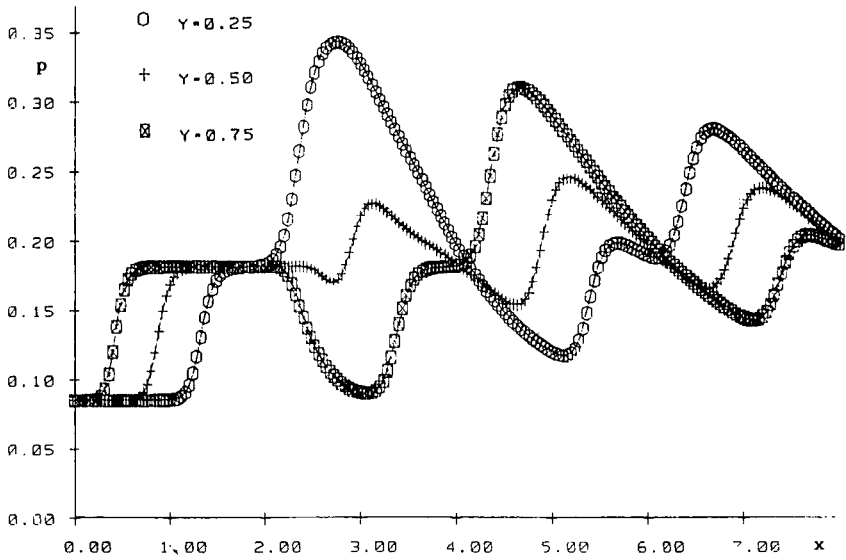


(5) *UP-1*

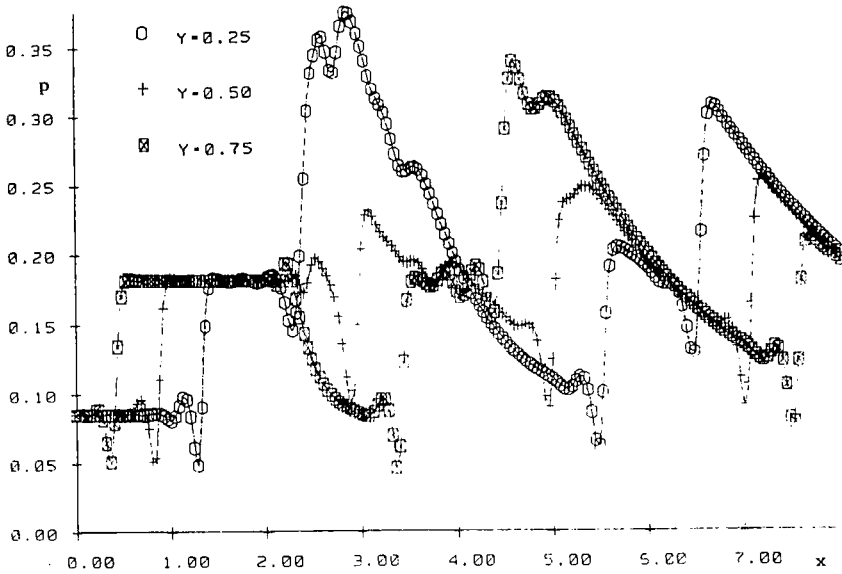


(6) *UP-2*

FIG. 8 (5)-(8) The pressure (a), density (b), Mach number (c) contours and pressure distributions calculated by *UP-1* and *UP-2*.



(1) UP-1



(6) UP-2

FIG. 8 (5)-(8)—continued.

We also have

$$\begin{aligned}\tau_{xx} &= \frac{\mu}{\text{Re}} \left(\frac{4}{3} \frac{\partial u}{\partial x} - \frac{2}{3} \frac{\partial v}{\partial y} \right), \\ \tau_{xy} &= \frac{\mu}{\text{Re}} \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right), \\ \tau_{yy} &= \frac{\mu}{\text{Re}} \left(-\frac{2}{3} \frac{\partial u}{\partial x} + \frac{4}{3} \frac{\partial v}{\partial y} \right), \\ f &= u\tau_{xx} + v\tau_{xy} + \frac{\mu}{\text{Re}} \frac{1}{\text{Pr}} \frac{\gamma}{\gamma - 1} \frac{\partial}{\partial x} \left(\frac{p}{\rho} \right), \\ g &= u\tau_{xy} + v\tau_{yy} + \frac{\mu}{\text{Re}} \frac{1}{\text{Pr}} \frac{\gamma}{\gamma - 1} \frac{\partial}{\partial y} \left(\frac{p}{\rho} \right).\end{aligned}$$

The dynamic viscosity μ is given by Sutherland's formula. The Reynolds number is Re , and the Prandtl number is $\text{Pr} = 0.72$. Under the generalized coordinate transformation $\xi = \xi(x, y)$ and $\eta = \eta(x, y)$, Eq. (6.2) can be written in a form that maintains the strong conservation law form as

$$\frac{\partial \hat{\mathbf{U}}}{\partial t} + \frac{\partial \hat{\mathbf{F}}}{\partial \xi} + \frac{\partial \hat{\mathbf{G}}}{\partial \eta} = \frac{1}{\text{Re}} \left(\frac{\partial \hat{\mathbf{F}}_v}{\partial \xi} + \frac{\partial \hat{\mathbf{G}}_v}{\partial \eta} \right), \quad (6.3)$$

where

$$\begin{aligned}\hat{\mathbf{U}} &= \mathbf{U}/J, \\ \hat{\mathbf{F}} &= (\xi_x F + \xi_y G)/J, \\ \hat{\mathbf{G}} &= (\eta_x F + \eta_y G)/J, \\ \hat{\mathbf{F}}_v &= (\xi_x F_v + \xi_y G_v)/J, \\ \hat{\mathbf{G}}_v &= (\eta_x F_v + \eta_y G_v)/J, \\ J &= \xi_x \eta_y - \xi_y \eta_x.\end{aligned}$$

The physical domain of interest is prescribed by three types of boundaries, namely an inflow, an outflow, and a solid wall. A scheme of the physical and computational domain is shown in Fig. 9. At the inflow boundary, the flow is supersonic and the flow conditions are given: the free-stream Mach number $M_\infty = 3$, the Reynolds number based on the freestream conditions and the diameter of cylinder $D \text{Re} = \rho_\infty V_\infty D / \mu_\infty = 10^5$, and the free-stream temperature $T_\infty = 980 \text{ K}$. At the outflow boundary, the flow is dominantly supersonic and a simple first-order extrapolation of all the flow variables is sufficient.

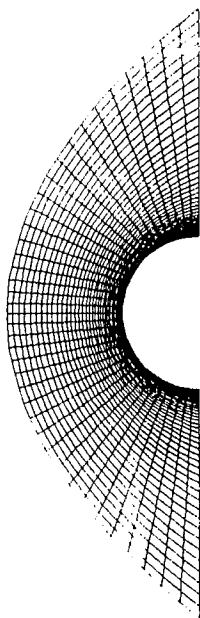


FIG. 9 Computational domain.

At the solid wall, no-slip boundary conditions $u = v = 0$ and an adiabatic wall boundary condition are considered. The surface pressure is obtained by solving the normal momentum equation. Since the Reynolds number is high in this case, the normal momentum equation can be approximated at the wall by $\partial p / \partial n = 0$, where n represents the direction normal to the body. The surface density is obtained by zeroth-order extrapolation to the surface.

2. Difference Method and Results

For simplicity, the mesh is generated using an algebraic grid generator that has the capability of clustering in the body normal direction to resolve the boundary layers, as well as to reach a specified outer boundary (Fig. 9).

Equation (6.2) is the same as Eq. (3.25), so the previous NND schemes solving (3.25) can be used here.

Figure 10 shows the pressure contours calculated with NND-1-5 and the implicit NND scheme. The velocity distributions on the symmetrical axis and the pressure distributions on the body surface are shown in Figs. 11 and 12, respectively.

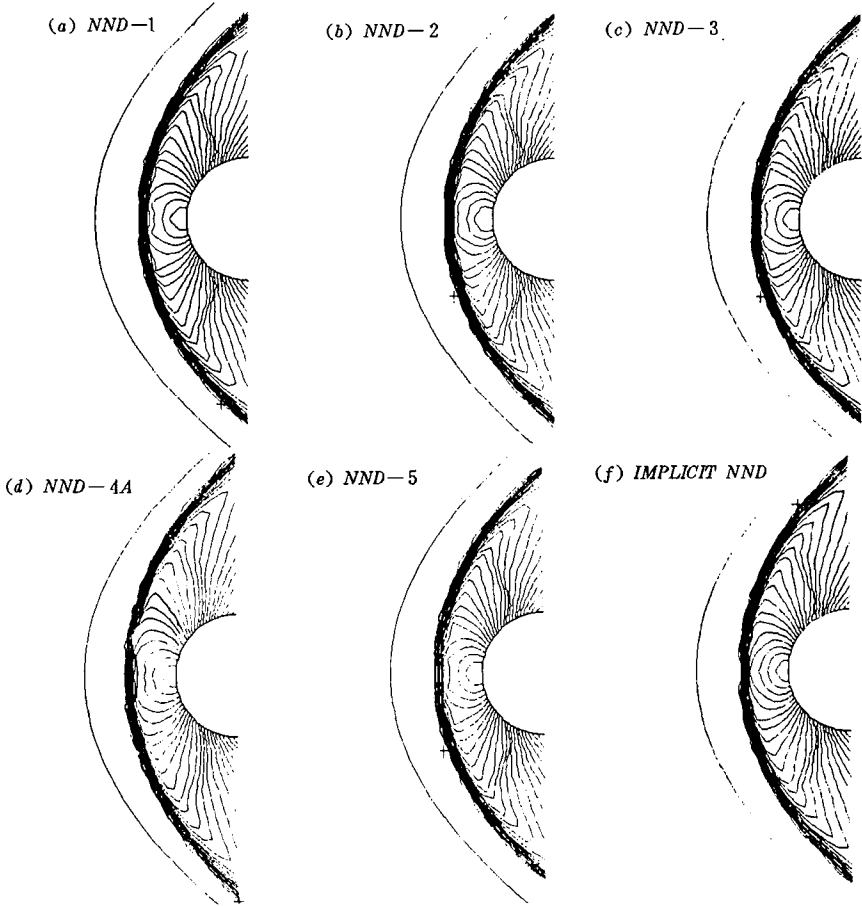


FIG. 10 Pressure contours calculated by NND-1-5 and implicit NND.

C. AXIAL SYMMETRIC FREE JET FLOWS (see Zheng and Zhang, 1989)

1. Nondimensional Equations and Boundary Conditions

Now we calculate the axial symmetric free jet with Euler and N-S equations. We can write the compressible equations in a cylindrical coordinate system (x, r, θ) (Fig. 13) as

$$\frac{\partial U}{\partial t} + \frac{\partial F}{\partial x} + \frac{\partial G}{\partial r} + H = \frac{\varepsilon}{\text{Re}_\infty} \left(\frac{\partial f}{\partial x} + \frac{1}{r} \frac{\partial g}{\partial r} + h \right), \quad (6.4)$$

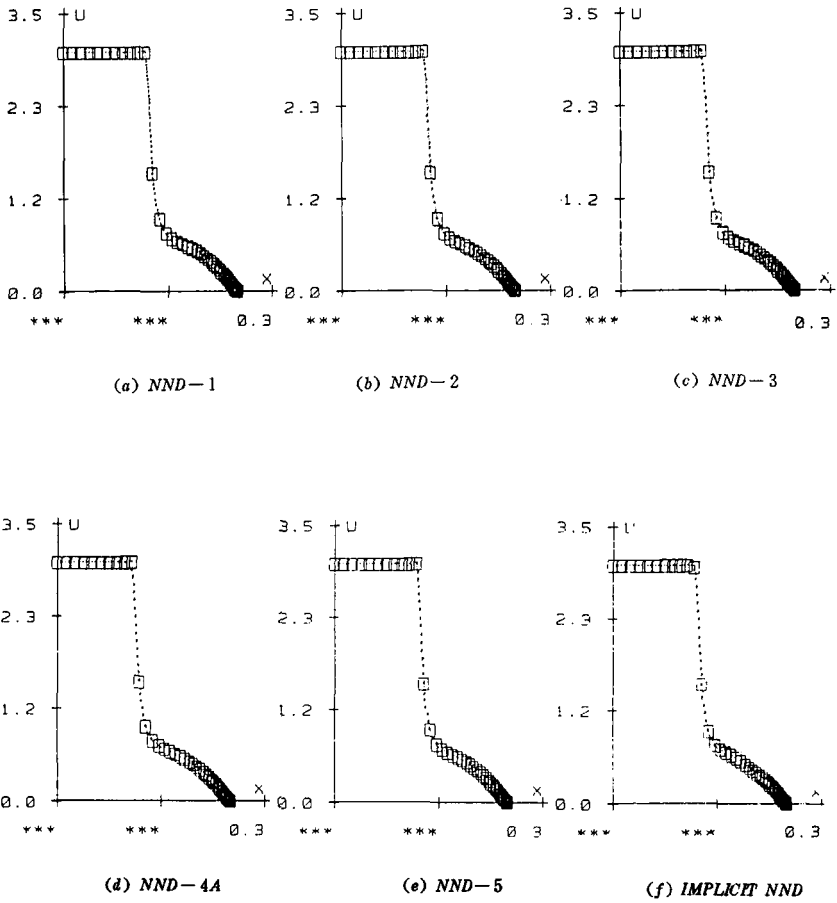
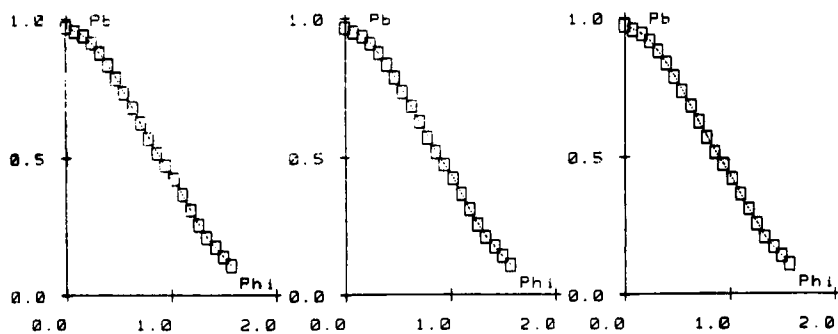


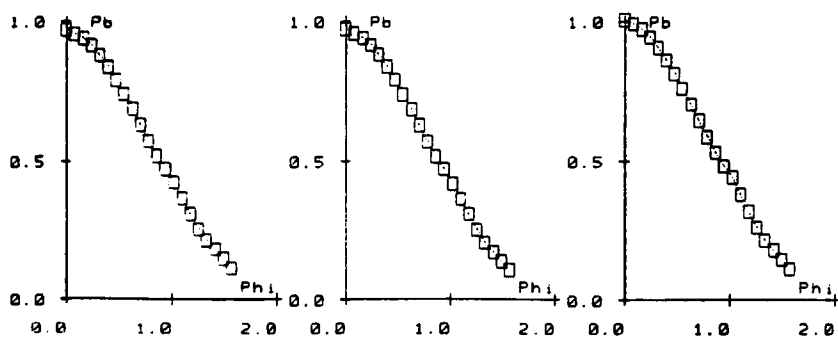
FIG. 11 Velocity distributions on the symmetrical axis calculated by NND-1-5 and implicit NND.



(a) NND-1

(b) NND-2

(c) NND-3



(d) NND-4A

(e) NND-5

(f) IMPLICIT NND

FIG. 12 Pressure distributions on the body surface calculated by NND-1-5 and implicit NND.

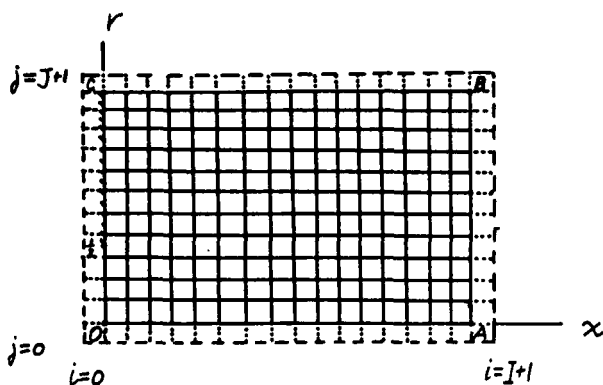


FIG. 13 The coordinate system and mesh.

where

$$\begin{aligned}
 U &= [\rho, \rho u, \rho v, e]^T, \\
 F &= [\rho u, \rho u^2 + p, \rho uv, (e + p)u]^T \\
 G &= [\rho v, \rho vu, \rho v^2 + p, (e + p)v]^T, \\
 H &= \frac{1}{r} [\rho v, \rho uv, \rho v^2, (e + p)v]^T, \\
 f &= \begin{bmatrix} 0 \\ \tau_{xx} \\ \tau_{xr} \\ u\tau_{xx} + v\tau_{xr} + q_x \end{bmatrix}, \\
 g &= r \begin{bmatrix} 0 \\ \tau_{xr} \\ \tau_{rr} \\ u\tau_{xr} + v\tau_{rr} + q_r \end{bmatrix}, \\
 h &= \frac{1}{r} [0, 0, -\tau_{\theta\theta}, 0]^T.
 \end{aligned}$$

For Euler equations, $\varepsilon = 0$, and for N-S equations, $\varepsilon = 1$. In Eq. (6.4), u , v represent the components of the velocity along the coordinate directions x , r , respectively. Equation (6.4) is expressed in nondimensional form,

with reference values for the velocity, pressure, density, temperature, viscosity, and total energy taken as $(p'_\infty/\rho'_\infty)^{1/2}$, p'_∞ , ρ'_∞ , T'_∞ , μ'_∞ , and p'_∞ , respectively, where the subscript ∞ represents the conditions of the external stream, which is at rest, and the dimensional variables are denoted by a '. The reference length is the exit diameter D' of the jet. The Reynolds number if $\text{Re}_\infty = (p'_\infty \rho'_\infty)^{1/2} D' / \mu'_\infty$.

We assume that in both external and internal regions the gas is perfect, and the equation of state is

$$p = \rho T.$$

For the viscous case considered here, the flow is laminar, and the viscosity μ is calculated by Sutherland's formula.

The nondimensional boundary conditions can be written as follows:

On the axis of symmetry ($r = 0$), we have $v = 0$. The outer boundary CB (Fig. 13) is set far away from the axis of symmetry. Here, the stream can be considered as at rest. So corresponding boundary conditions are

$$\rho = 1, \quad p = 1, \quad u = v = 0.$$

The left boundary consists of two parts. On the body surface, we have

$$u = 0$$

for the inviscid case, and

$$u = v = 0, \quad T = T_w$$

for the viscous flow, where T_w is the surface temperature. On the exit of the jet, the following conditions can be given:

$$\begin{cases} p = p_0 \left(\frac{\gamma + 1}{2} \right)^{-\gamma/\gamma-1}, \\ \rho = \rho_0 \left(\frac{\gamma + 1}{2} \right)^{-1/\gamma-1}, \\ u = \left(\gamma \frac{p}{\rho} \right)^{1/2}, \\ v = 0, \end{cases}$$

where the stagnation condition is denoted by subscript 0. Supposing the right boundary is set far away from the exit of the jet, so zero-gradient boundary conditions can be used:

$$\frac{\partial \varphi}{\partial x} = 0,$$

where $\varphi = u, v, p, \rho$.

2. Difference Equations

The mesh used for the simulation is shown in Fig. 13. Using the NND-1 scheme, the following difference equations can be obtained from Eq. (6.4):

$$U_{ij}^{n+1} = U_{ij}^{n+1} + \frac{\varepsilon \Delta t}{\text{Re}_\infty} \left[\left(\frac{\partial f}{\partial x} \right)_{ij}^n + \left(\frac{1}{r} \frac{\partial g}{\partial r} \right)_{ij}^n + h_{ij}^n \right] - H_{ij}^n \Delta t,$$

where

$$U_{ij}^{n+1} = L_x(\tfrac{1}{2} \Delta t) L_r(\Delta t) L_x(\tfrac{1}{2} \Delta t) U_{ij}^n,$$

$$L_x(\Delta t) U_{ij}^n = U_{ij}^n - (\Delta t / \Delta x) (X_{i+1/2j}^n - X_{i-1/2j}^n),$$

$$X_{i+1/2j}^n = F_{i+1/2jL}^+ + F_{i+1/2jR}^-,$$

$$F_{i+1/2jL}^+ = F_{ij}^+ + \tfrac{1}{2} \min\text{mod}(\Delta F_{i-1/2j}^+, \Delta F_{i+1/2j}^+),$$

$$F_{i+1/2jR}^- = F_{i+1j}^- - \tfrac{1}{2} \min\text{mod}(\Delta F_{i+1/2j}^-, \Delta F_{i+3/2j}^-),$$

$$L_r(\Delta t) U_{ij}^n = U_{ij}^n - (\Delta t / \Delta r) (Y_{ij+1/2}^n - Y_{ij-1/2}^n),$$

$$Y_{ij+1/2}^n = G_{ij+1/2L}^+ + G_{ij+1/2R}^-,$$

$$G_{ij+1/2L}^+ = G_{ij}^+ + \tfrac{1}{2} \min\text{mod}(\Delta G_{ij-1/2}^+, \Delta G_{ij+1/2}^+),$$

$$G_{ij+1/2R}^- = G_{ij+1}^- - \tfrac{1}{2} \min\text{mod}(\Delta G_{ij+1/2}^-, \Delta G_{ij+3/2}^-),$$

and $(\partial f / \partial x)_{ij}^n$, $(\partial g / \partial x)_{ij}^n$ can be calculated using second-order central difference scheme.

3. Numerical Results

For the axial symmetric free jet simulated here, the ratio of the exit total pressure p'_{0j} to the external pressure p'_∞ is 50; the exit Mach number $M = 1.0$. The total enthalpy is the same for both external and internal stream ($H'_j = H'_\infty$). The ratio of specific heats is 1.4 for both external and internal streams. We assume that $T_w = T'_\infty = 300$ K and

$$\text{Re}_{\text{jet}} = \frac{\rho'_{ij} u'_{ij} D'}{\mu'_{ij}} = \sqrt{\gamma p_{ij} \rho_{ij}} \frac{1}{\mu_{ij}} \text{Re}_\infty = 10^5.$$

The mesh system is 60×60 . The time step Δt is determined by following

stability conditions:

$$\Delta t = \frac{2}{3} \min(\Delta t_x, \Delta t_r),$$

$$\Delta t_x = \frac{\Delta x}{\left\{ |u| + \alpha + \frac{1}{\rho \text{Re}_\infty} \left[\frac{2\gamma\mu}{\text{Pr} \Delta x} + \left(\frac{2}{3} \mu^2 \right)^{1/2} \frac{1}{\Delta r} \right] \right\}_{\max}},$$

$$\Delta t_r = \frac{\Delta r}{\left\{ |v| + \alpha + \frac{1}{\rho \text{Re}_\infty} \left[\frac{2\gamma\mu}{\text{Pr} \Delta r} + \left(\frac{2}{3} \mu^2 \right)^{1/2} \frac{1}{\Delta x} \right] \right\}_{\max}},$$

For air, the Prandtl number $\text{Pr} = 0.72$. Figure 14(a, b) give the density and pressure contours, respectively, for the axial symmetric free jet with Euler and N-S equations. The existence of the drum shock and Mach disc is clearly

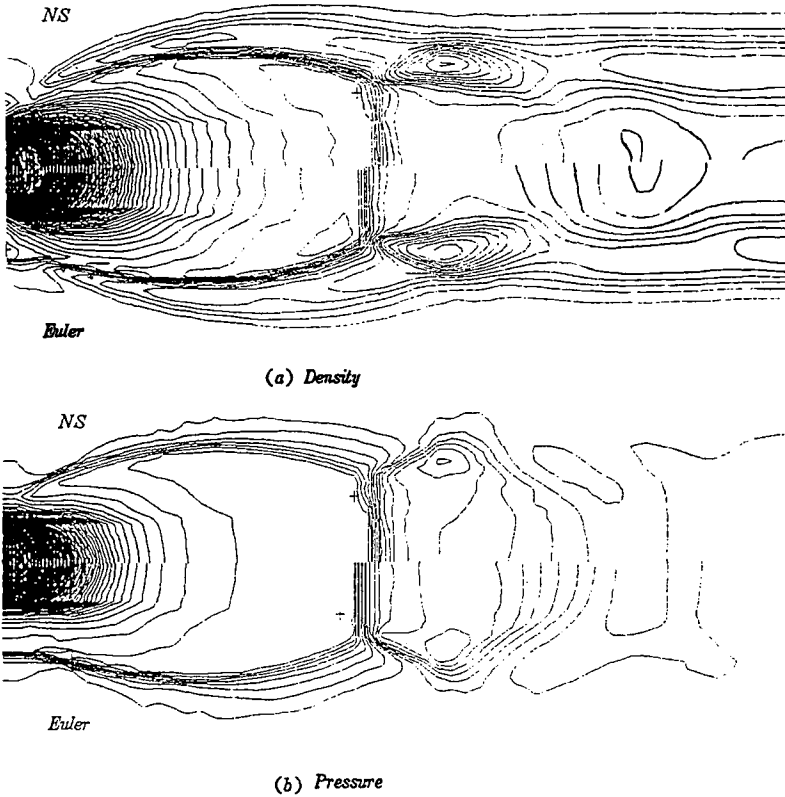


FIG. 14 Density and pressure contours calculated by NND-1.

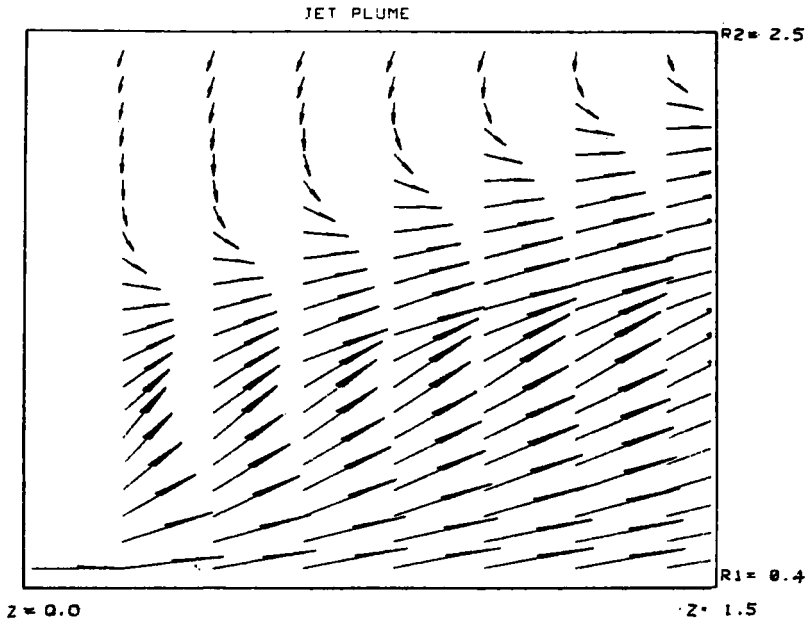


FIG. 15 Velocity vectors near the jet.

visible, and also a density jump across the contact discontinuity. Figure 15 shows the velocity vectors near the jet boundary with N-S equations. The viscous effects on the location of Mach disc and the flow pattern near the jet boundary are obvious. The flow pattern obtained by experiment is given in Teshima (1984). The location of the Mach disc and the diameter of the drum shock satisfactorily agree with the present calculation with N-S equations. Figure 16 gives the calculated axial pressure, density, temperature, and Mach number distributions using the Euler equations. We note that behind the Mach disc there appear no oscillations. Figure 17 and 18 give the comparison of calculated axial and radial density, pressure, and Mach number distributions using N-S equations with the results using Euler equations.

D. HYPERSONIC FLOW AROUND A BLUNT BODY (see Ye, 1989 and Shen, 1989)

The computation of 3D hypersonic flow around blunt bodies is of increasing interest for the future development of spacecrafts and hypersonic

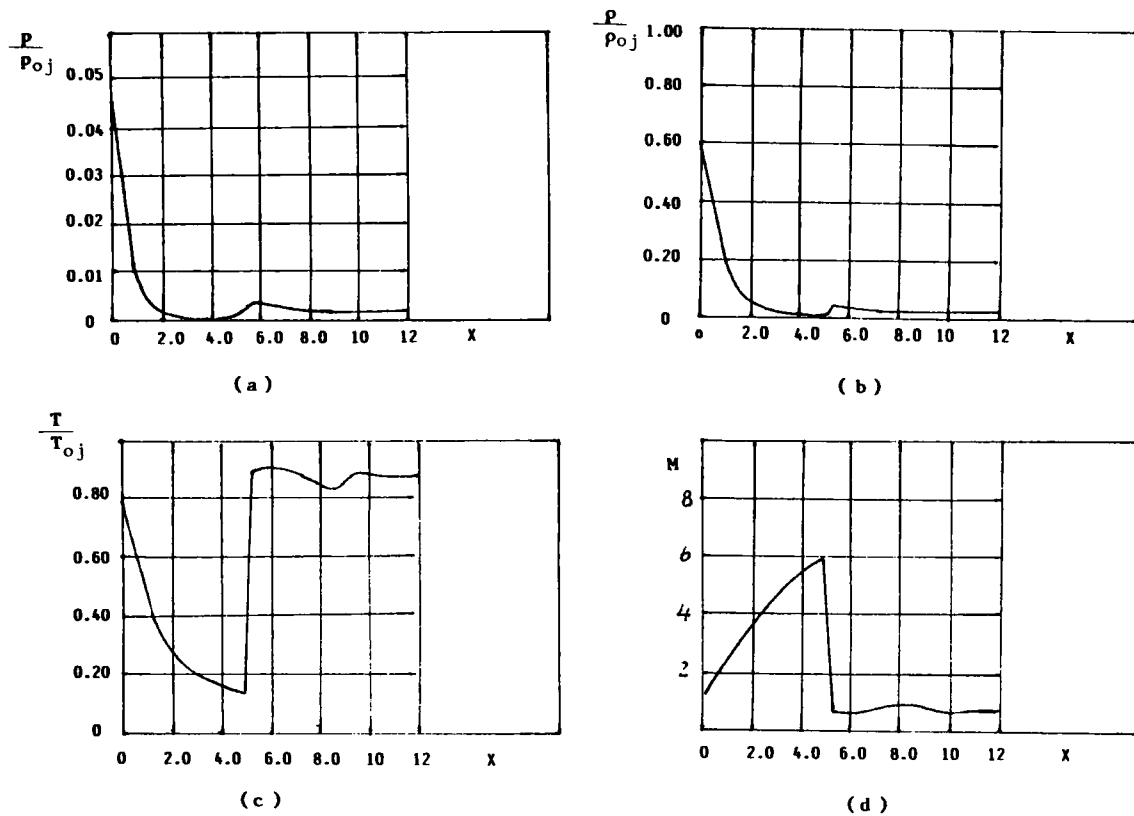
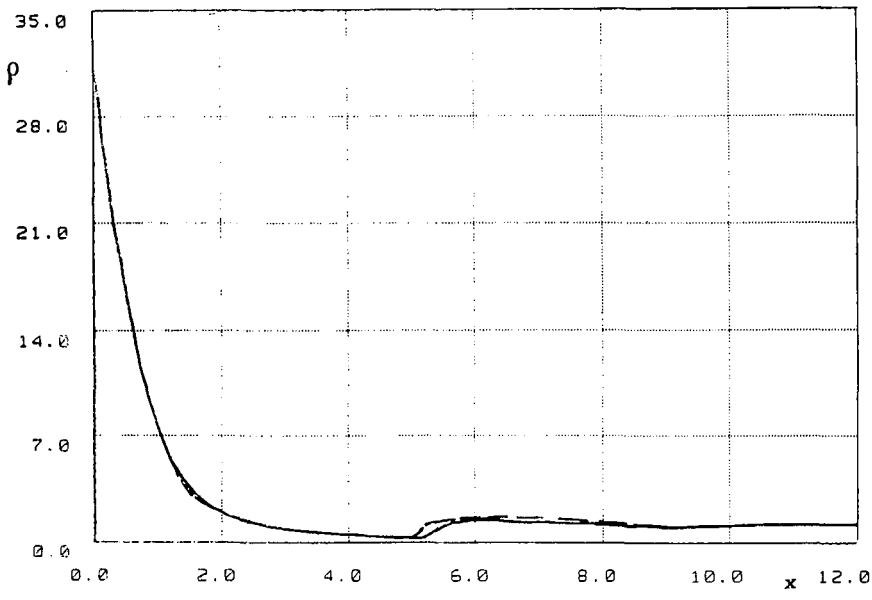
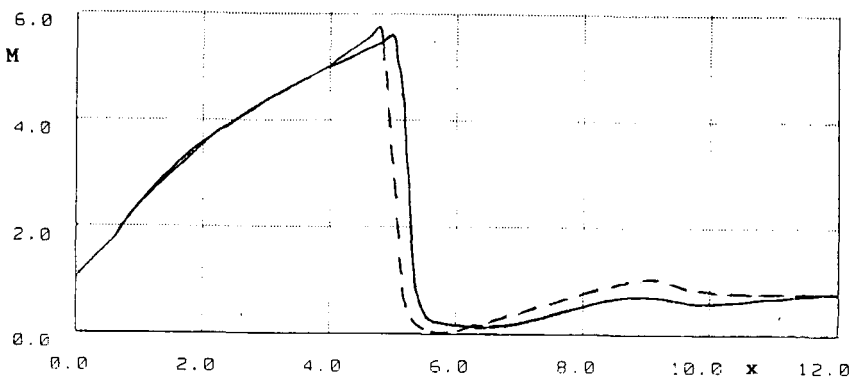


FIG. 16 Calculated pressure (a), density (b), temperature (c) and Mach number (d) distribution (— present).



(a)



(b)

FIG. 17 Calculated axial density (a) and Mach number (b) distributions. — NS, --- Euler.

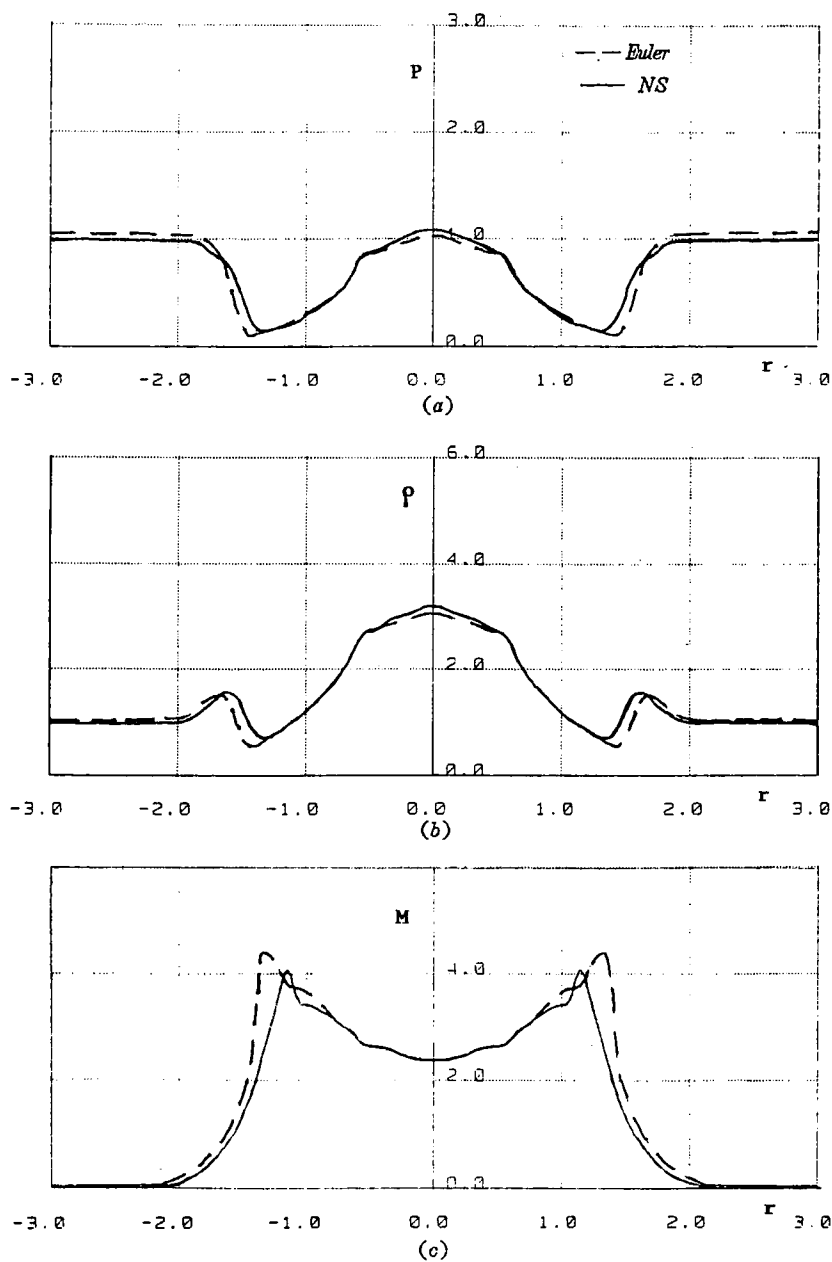


FIG. 18 Calculated radial pressure (a), density (b) and Mach number (c) distributions.

airplanes. The flow around blunt bodies is characterized by strong shock waves, expansions, and embedded subsonic regions. To attack such flow problems, the Euler equations and Navier-Stokes equations are solved.

1. Governing Equations

The three-dimensional flow of the compressible gas is described by the time-dependent Euler or Navier-Stokes equations for the inviscid flow or the viscous flow. The conservative nondimensional equations are formulated in a curvilinear body fitted time-dependent coordinate system with

$$\begin{cases} \tau = t, \\ \xi = \xi(t, x, y, z), \\ \eta = \eta(t, x, y, z), \\ \zeta = \zeta(t, x, y, z). \end{cases}$$

Then, the governing equations read

$$\frac{\partial \mathbf{U}}{\partial \tau} + \frac{\partial E_1}{\partial \xi} + \frac{\partial F_1}{\partial \eta} + \frac{\partial G_1}{\partial \zeta} = \varepsilon \left(\frac{\partial E_2}{\partial \xi} + \frac{\partial F_2}{\partial \eta} + \frac{\partial G_2}{\partial \zeta} \right), \quad (6.5)$$

where $\varepsilon = 1$ for N-S equations, and $\varepsilon = 0$ for Euler equations, and

$$\mathbf{U} = J\hat{\mathbf{U}},$$

$$E_1 = J(\xi_t \hat{\mathbf{U}} + \xi_x \hat{E}_1 + \xi_y \hat{F}_1 + \xi_z \hat{G}_1),$$

$$E_2 = J(\xi_x \hat{E}_2 + \xi_y \hat{F}_2 + \xi_z \hat{G}_2),$$

$$F_1 = J(\eta_t \hat{\mathbf{U}} + \eta_x \hat{E}_1 + \eta_y \hat{F}_1 + \eta_z \hat{G}_1),$$

$$F_2 = J(\eta_x \hat{E}_2 + \eta_y \hat{F}_2 + \eta_z \hat{G}_2),$$

$$G_1 = J(\zeta_t \hat{\mathbf{U}} + \zeta_x \hat{E}_1 + \zeta_y \hat{F}_1 + \zeta_z \hat{G}_1),$$

$$G_2 = J(\zeta_x \hat{E}_2 + \zeta_y \hat{F}_2 + \zeta_z \hat{G}_2),$$

$$\xi_x = (y_\eta z_\zeta - y_\zeta z_\eta)/J,$$

$$\xi_y = (z_\eta x_\zeta - x_\eta z_\zeta)/J,$$

$$\xi_z = (x_\eta y_\zeta - y_\eta x_\zeta)/J,$$

$$\eta_x = (z_\xi y_\zeta - y_\xi z_\zeta)/J,$$

$$\eta_y = (x_\xi z_\zeta - x_\zeta z_\xi)/J,$$

$$\eta_z = (y_\xi x_\zeta - x_\xi y_\zeta)/J,$$

$$\zeta_x = (y_\xi z_\eta - z_\xi y_\eta)/J,$$

$$\zeta_y = (x_\eta z_\xi - z_\eta x_\xi)/J,$$

$$\zeta_z = (x_\xi y_\eta - y_\xi x_\eta)/J,$$

$$\xi_t = -x_\tau \xi_x - y_\tau \xi_y - z_\tau \xi_z,$$

$$\eta_t = -x_\tau \eta_x - y_\tau \eta_y - z_\tau \eta_z,$$

$$\zeta_t = -x_\tau \zeta_x - y_\tau \zeta_y - z_\tau \zeta_z,$$

$$1/J = \partial(\xi, \eta, \zeta)/\partial(x, y, z).$$

In the preceding, the $\tilde{\mathbf{U}}$, $\hat{E}_1$, $\hat{E}_2$, $\hat{F}_1$, $\hat{F}_2$, $\hat{G}_1$, $\hat{G}_2$ can be expressed as

$$\tilde{\mathbf{U}} = [\rho, \rho u, \rho v, \rho w, e]^T,$$

$$\hat{E}_1 = [\rho u, \rho u^2 + p, \rho uv, \rho uw, eu + pu]^T,$$

$$\hat{E}_2 = \frac{1}{\text{Re}} \begin{bmatrix} 0 \\ \tau_{xx} \\ \tau_{xy} \\ \tau_{xz} \\ \tau_{xx}u + \tau_{xy}v + \tau_{xz}w + k \frac{\partial T}{\partial x} \end{bmatrix},$$

$$\hat{F}_1 = [\rho v, \rho uv, \rho v^2 + p, \rho vw, (e + p)v]^T,$$

$$\hat{F}_2 = \frac{1}{\text{Re}} \begin{bmatrix} 0 \\ \tau_{xy} \\ \tau_{yy} \\ \tau_{yz} \\ \tau_{xy}u + \tau_{yy}v + \tau_{yz}w + k \frac{\partial T}{\partial y} \end{bmatrix},$$

$$\hat{G}_1 = [\rho w, \rho wu, \rho wv, \rho w^2 + p, (e + p)w]^T,$$

$$\hat{G}_2 = \frac{1}{\text{Re}} \begin{bmatrix} 0 \\ \tau_{xz} \\ \tau_{yz} \\ \tau_{zz} \\ u\tau_{xz} + v\tau_{yz} + w\tau_{zz} + k \frac{\partial T}{\partial z} \end{bmatrix},$$

and

$$\tau_{xx} = 2\mu \frac{\partial u}{\partial x} - \frac{2}{3} \mu \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} \right),$$

$$\tau_{yy} = 2\mu \frac{\partial v}{\partial y} - \frac{2}{3} \mu \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} \right),$$

$$\tau_{zz} = 2\mu \frac{\partial w}{\partial z} - \frac{2}{3} \mu \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} \right),$$

$$\tau_{xy} = \mu \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \right),$$

$$\tau_{yz} = \mu \left(\frac{\partial w}{\partial y} + \frac{\partial v}{\partial z} \right),$$

$$\tau_{zx} = \mu \left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x} \right),$$

where u , v , and w are the velocity components in the Cartesian system (x, y, z) .

The coordinate system (ξ, η, ζ) is chosen nearly orthogonal, with the ζ -coordinate normal to the body surface. The ξ -coordinates and η -coordinates are the streamwise and the circumferential directions, respectively. The interior mesh is created with the body and the bow shock wave as two boundaries, and a fine clustering mesh spacing is used near the body surface in order to resolve viscous effects. The calculated formation for interior mesh in Cartesian system is

$$\begin{cases} x = x_w + \alpha(x_s - x_w), \\ y = y_w + \alpha(y_s - y_w), \\ z = z_w + \alpha(z_s - z_w) \end{cases}$$

and

$$\alpha = 1 + \beta \left[1 - \left(\frac{\beta + 1}{\beta - 1} \right)^{1-b} \right] / \left[1 + \left(\frac{\beta + 1}{\beta - 1} \right)^{1-b} \right],$$

$$b = \frac{K - 1}{K_{\max} - 1},$$

where w and s represent the bow shock and the body surface, β is the clustering factor, and K is indexing of the computational mesh in the ζ -direction.

2. Boundary Conditions

As an example, the flow around a double-ellipsoidal body was calculated. The geometry is shown in Fig. 19. In the present numerical procedure, the bow shock wave is treated as a discontinuous surface, and the shock-fitting scheme is introduced for the computational efficiency. Rankine-Hugoniot relations and one characteristic relation derived from Eq. (6.5) are applied to determine the flow just behind the shock wave, and the shock speed. No-slip condition requires $u = v = w = 0$ on the body surface for the viscous flow. The surface pressure p is obtained from the zero normal pressure gradient. The surface temperature T_w is held constant. Then, the surface density ρ is obtained from the equation of state. For the inviscid flow, the body surface boundary condition requires that the velocity component normal to the body be zero; four characteristic relations derived from Euler equations are used. On the outflow plane, a linear extrapolation is imposed. On the symmetrical plane, the symmetry conditions are enforced.

The initial flow fields are determined by assuming the suitable shock-wave shapes with Rankine-Hugoniot relations, and using modified Newtonian surface pressure distribution along the body surface. Between the body and the shock wave, a linear variation of the flow variables is assumed.

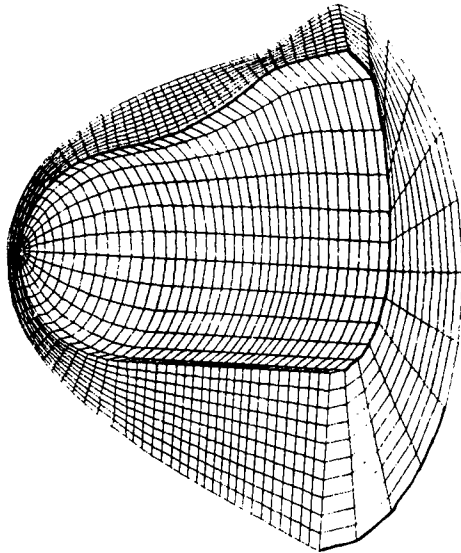


FIG. 19 Geometry and grid.

3. Difference Method and Results

Since Eq. (6.5) is of the same form as Eq. (3.25), time-splitting scheme presented there can be applied in the present case. For simplicity the NND-1 scheme is used here.

a. Inviscid Solution

The inviscid solution has been obtained at following free stream conditions: $M_\infty = 10$; $\alpha = -5^\circ$.

The domain of the calculation and mesh are shown in Fig. 19. In Fig. 20, the lines of constant pressure and Mach number on the symmetry plane and the body surface are given. The $41 \times 17 \times 11$ (streamwise $\times$ circumferential $\times$ surface normal) grid is used.

b. Viscous Solution

Here, the $45 \times 19 \times 26$ (streamwise $\times$ circumferential $\times$ surface normal) grid and $\beta = 1.03$ are used. In Fig. 21, the lines of constant pressure and temperature are shown.

E. HYPERSONIC FLOW AROUND SPACE-SHUTTLE-LIKE GEOMETRY

Now we study the hypersonic flow around space-shuttle-like geometry, which is modified geometry for complete space shuttle orbiter and is called as "QUICK geometry model" (see Weilmuenster and Hamilton, 1983) (Fig. 22).

1. Governing Equations and Difference Method

The Euler equations are used to solve the fore- and afterbody flowfield over this geometry. In the forebody region (blunt-nose region), the time-dependent blunt body code developed using the NND-1 scheme is used to obtain the blunt-nose solution. The flow is supersonic in the afterbody flowfield when the angle of attack is not very large, so the space-marching method is applied to solve steady Euler equations. Here, we discuss only the calculation of the afterbody flowfield.

In a nearly orthogonal curvilinear body-fitted coordinate system with

$$\begin{cases} \xi = \xi(x), \\ \eta = \eta(x, y, z), \\ \zeta = \zeta(x, y, z), \end{cases}$$

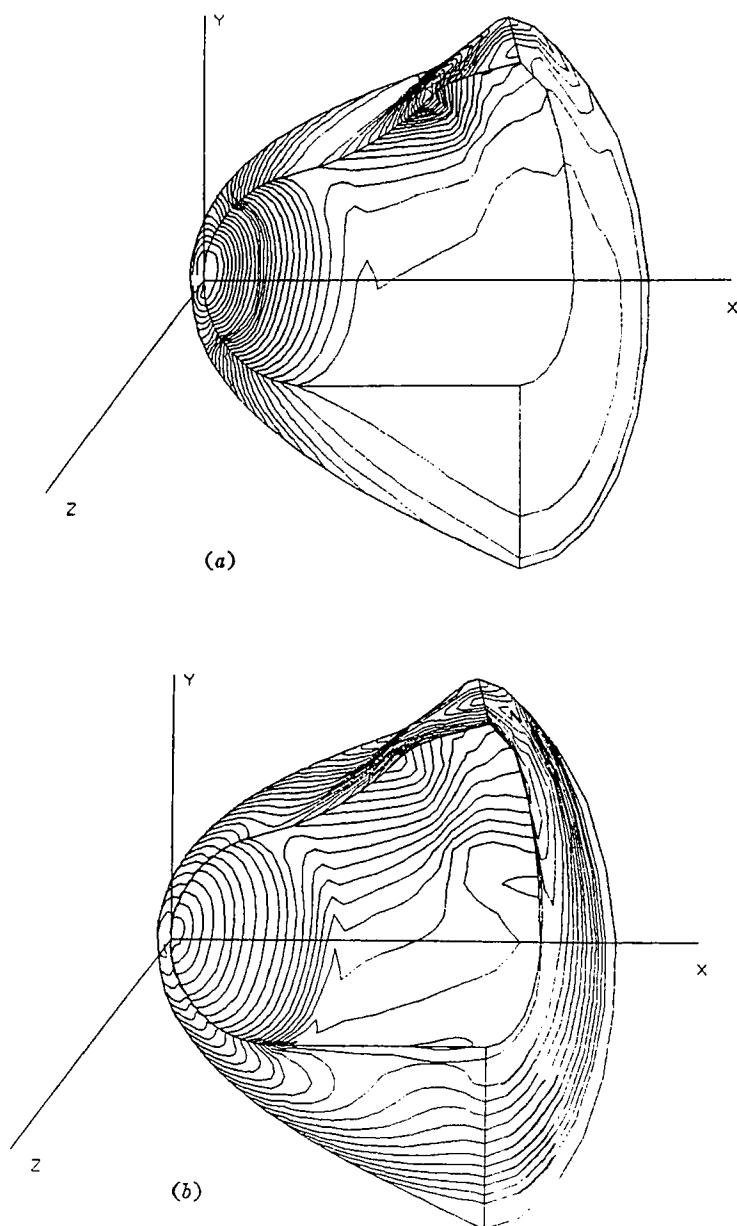


FIG. 20 Pressure (a) and Mach number (b) contours calculated using Euler equations.

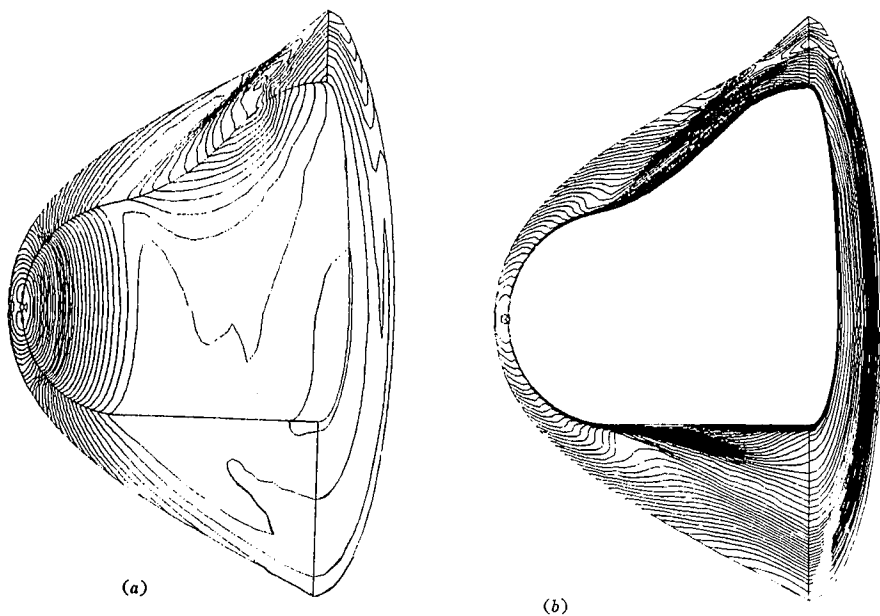


FIG. 21 Pressure (a) and temperature (b) contours calculated using NS equations.

where x, y, z are Cartesian coordinates and ξ, η, ζ are the coordinates of the streamwise, circumferential and wall-normal directions, respectively, the three-dimensional compressible steady Euler equations can be written in dimensionless form as

$$\frac{\partial \tilde{E}}{\partial \xi} + \frac{\partial \tilde{F}}{\partial \eta} + \frac{\partial \tilde{G}}{\partial \zeta} = 0, \quad (6.6)$$

where

$$\tilde{E} = \xi_x E / J,$$

$$\tilde{F} = (\eta_x E + \eta_y F + \eta_z G) / J,$$

$$\tilde{G} = (\zeta_x E + \zeta_y F + \zeta_z G) / J,$$

$$F = [\rho v, \rho uv, \rho v^2 + p, \rho vw, ev + pv]^T,$$

$$G = [\rho w, \rho uw, \rho vw, \rho w^2 + p, ew + pw]^T,$$

$$J = \partial(\xi, \eta, \zeta) / \partial(x, y, z).$$

Since Eq. (6.6) is hyperbolic, and $\tilde{F} = \tilde{F}(\tilde{E})$, $\tilde{G} = \tilde{G}(\tilde{E})$, the NND-4(b) scheme can be used.

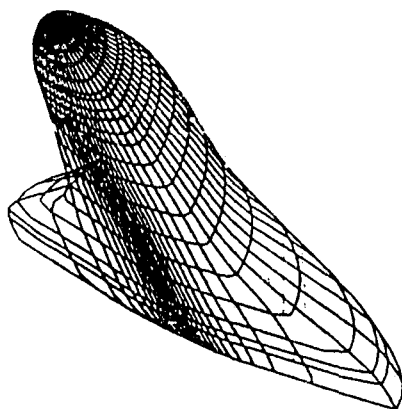


FIG. 22 QUICK geometry model.

2. Boundary Conditions and Results

Using the tangential-flow condition on body surface, and four characteristic relations derived from Euler equations, all variables on the body can be determined. At the shock wave, the Rankine-Hugoniot relations and one characteristic relation can be used to determine the shock-wave shape and all variables. The initial profile can be given by the time-dependent blunt-body code with the NND-1 scheme. The inviscid solution has been obtained at the following free-stream condition:

$$M_{\infty} = 10, \quad \alpha = 5^{\circ}.$$

In Fig. 23, the pressure and Mach number contours on the leeward side are given. Figure 24 shows the pressure and Mach number contours on windward side. The pressure, density, and Mach number contours in the cross sections at $x = 24$ and 26 are shown in Fig. 25. In Fig. 26, the pressure distributions on three meridional surfaces ($\varphi = 83.57^{\circ}$, 90° , and 96.43°) are given, and the pressure distribution on the cross sections ($x = 23, 24, 25$, and 26) are shown in Fig. 27. Figure 28 shows the pressure and density contours on the meridional surface $\varphi = 90^{\circ}$. The preceding results show that the bow shock and flowfields are accurately calculated, and that the canopy shocks are accurately captured with no oscillations. The capabilities of capturing shock and other discontinuities with NND schemes is satisfactory.

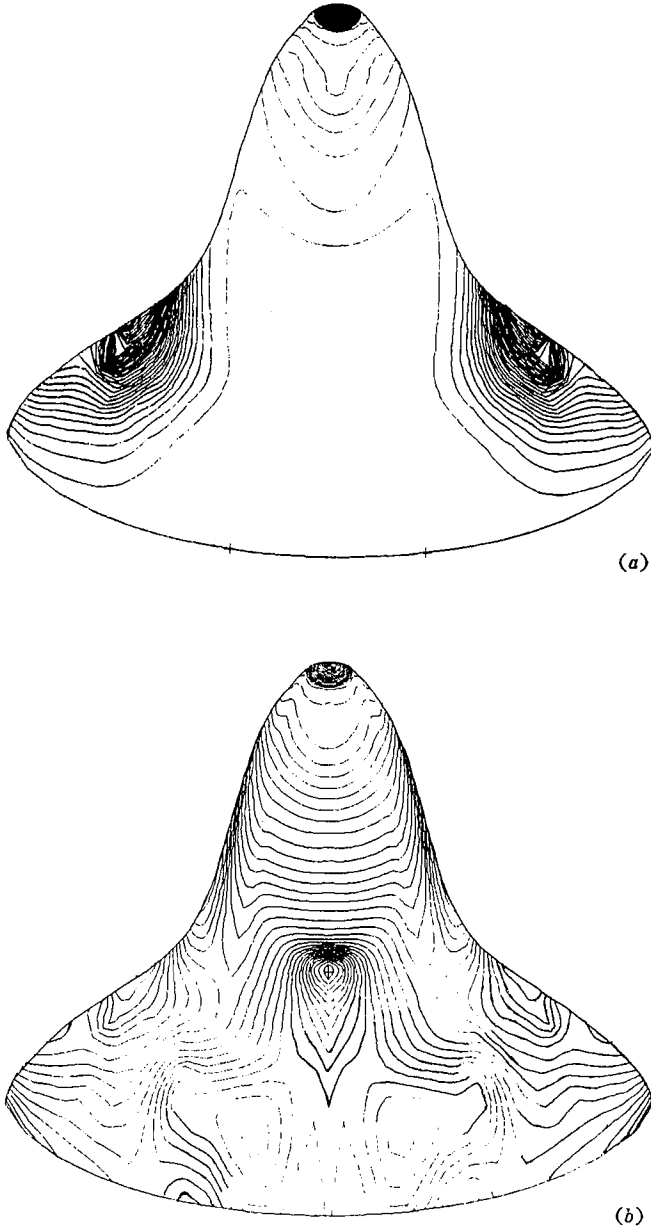


FIG. 23 The pressure (a) and Mach number (b) contours on the leeward side.

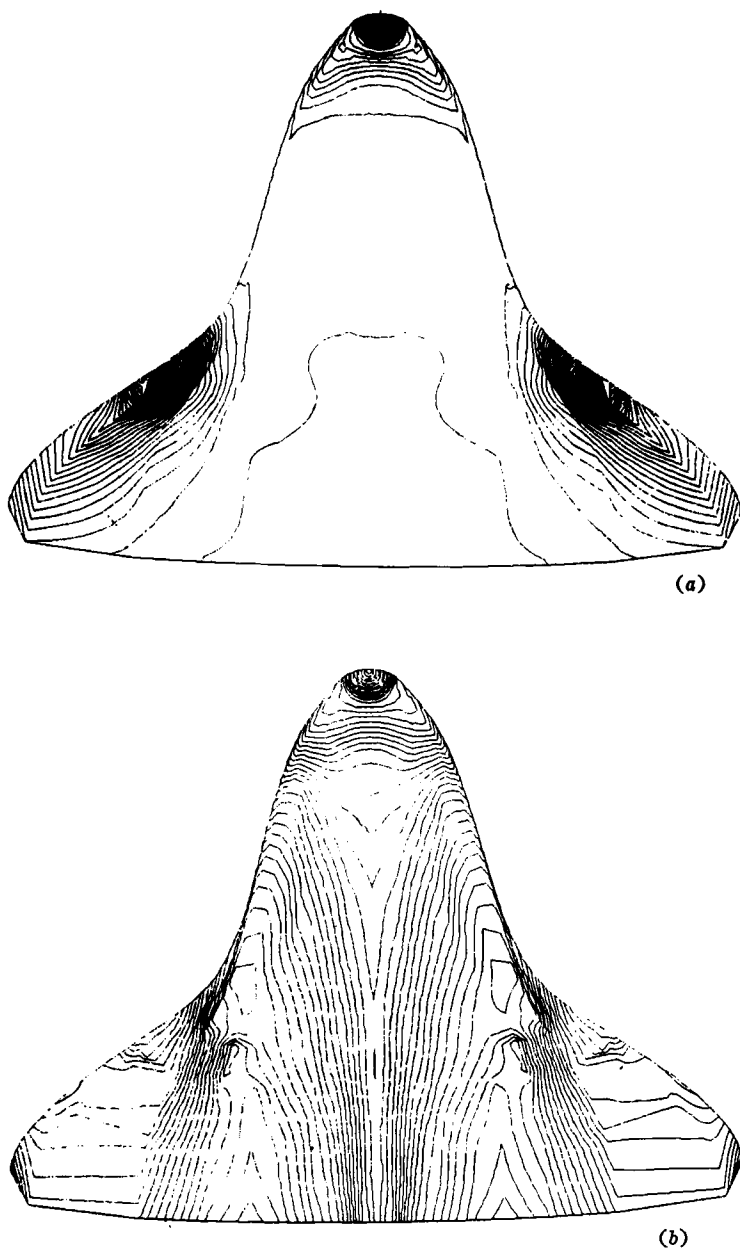


FIG. 24 The pressure (a) and Mach number (b) contours on the windward side.

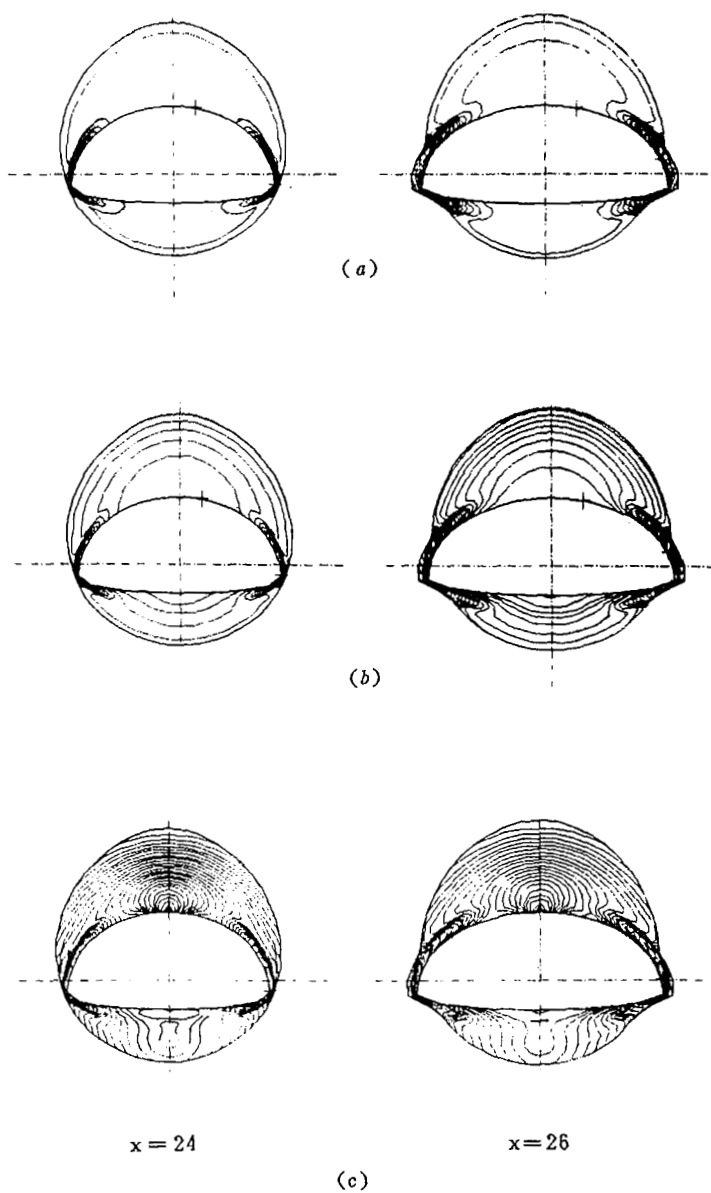


FIG. 25 Pressure (a), density (b) and Mach number (c) contours on the cross sections at $x = 24, 26$.

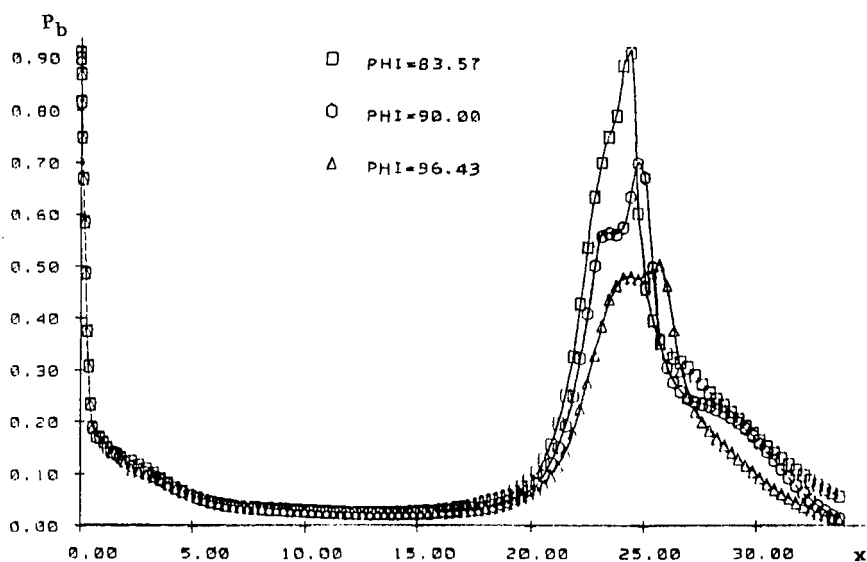


FIG. 26. Pressure distributions on the three dimensional surfaces.

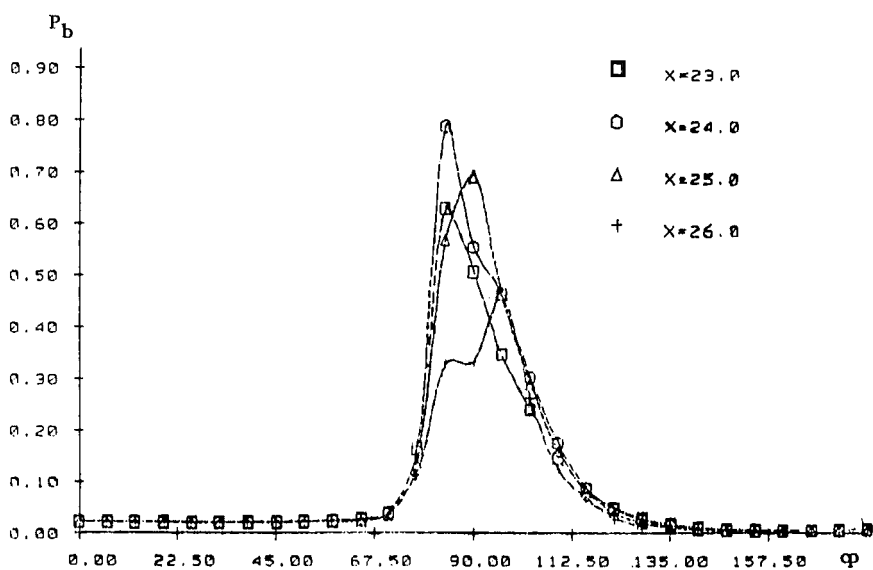
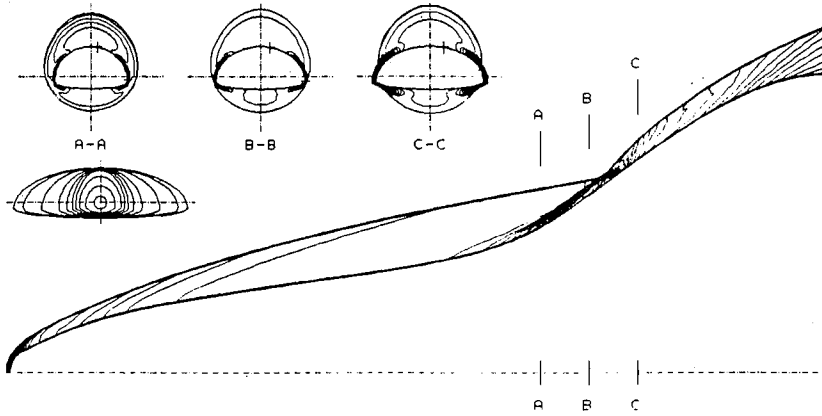
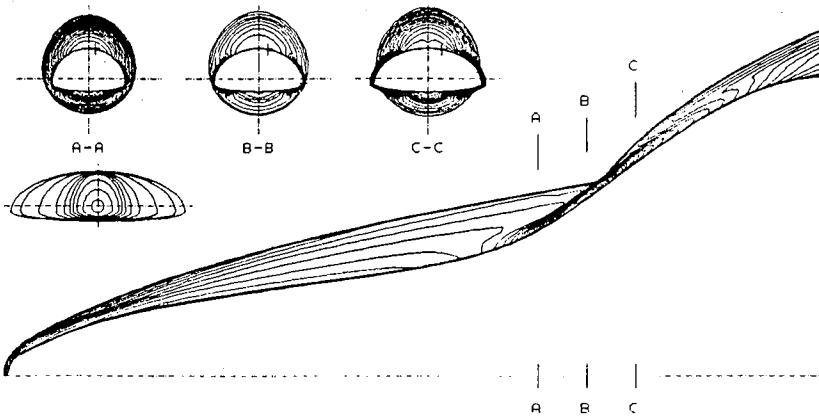


FIG. 27. Pressure distributions on the cross sections.



(a)



(b)

FIG. 28. Pressure (a) and density (b) contours on the meridional surface $\varphi = 90^\circ$.

VII. Concluding Remarks

Through a study for one-dimensional N-S equations, it was found that the spurious oscillations occurring near shock wave with finite-difference equations are related to the dispersion term in the corresponding modified differential equations. If the sign of the dispersion coefficient ν_3 is properly

adjusted, that is, the sign changes across shock wave, $v_3 > 0$ upstream, and $v_3 < 0$ downstream, the undesirable oscillations can be totally suppressed. Based on this finding, the semidiscretized nonoscillatory, containing no free parameters and dissipative scheme is developed, and five explicit NND schemes (NND 1–5) and an implicit NND scheme are given. We can prove that all NND schemes possess the TVD property.

It is very interesting to note the following facts:

- (1) The expression of NND-1 scheme is formally just the same as the Osher–Chakravarthy scheme based on the second-order upwind method except for the delimiter parameters β , and NND-1 is of second-order accuracy in regions of steep gradients as well as smooth regions.
- (2) NND-2 is just the Harten TVD scheme.
- (3) NND-4 is the TVD scheme that can be looked upon as an extension of MacCormack's predictor—corrector scheme.
- (4) From the consideration of the amount of numerical work required, using NND-1, NND-4, NND-5, or implicit NND schemes is suitable.

In general, we have seen that the NND schemes developed so far are very flexible, and are easily adapted to solve different kinds of flow problems, either in the time-dependent manner or in the direct treatment of the steady case, e.g., marching in the region of supersonic flow. Besides from the derivation of the scheme at its inception, this kind of consideration may be applied to the calculations of flow region where we have contact discontinuities or shear flows with steep gradients of the flow parameters, as already exemplified here to a certain extent by our numerical simulations. From the preceding calculated results, we have gained confidence in the use of the NND schemes suggested here. The distinguished feature of these schemes is the capability of capturing shock and other contact discontinuities, as exemplified by the preceding graphical representations. The schemes possess good stability characteristics and converged accuracy; this is essential to any high-shock resolution scheme. The present form seems to be the simplest; meanwhile, the amount of numerical work is much reduced in comparison with some other high-resolution TVD schemes.

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